Type MS · Where lines cross: two routes and comparing coefficients

Answer 1 of 20 · Question 11 part b · 7 marks

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11 a i	$\left(\overrightarrow{AN}=\right)\overrightarrow{AO}+\overrightarrow{ON}$ or $\left(\overrightarrow{AN}=\right)\overrightarrow{AO}+\frac{3}{4}\overrightarrow{OB}=-\mathbf{a}+\frac{3}{4}\mathbf{b}$	M1 A1
ii	$\left(\overrightarrow{BM}=\overrightarrow{BO}+\overrightarrow{OM}=\right)-\mathbf{b}+\frac{1}{2}\mathbf{a}$	B1 (A1 on ePen) (3)
b	$\left(\overrightarrow{AX}=\lambda\left(\pm\left(\text{their }\overrightarrow{AN}\right)\right)\right)=\lambda\left(\pm"-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)$	B1 ft
	$\left(\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(\pm\left(\text{their }\overrightarrow{MB}\right)\right)=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm"-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)$	M1
	Equating components gives $-\lambda=-\frac{1}{2}-\frac{1}{2}\mu$	M1
	and $\frac{3}{4}\lambda=\mu$	M1
	$-\lambda=-\frac{1}{2}-\frac{1}{2}\left(\frac{3}{4}\lambda\right)\Rightarrow\lambda=\frac{4}{5}$ *or* $\mu=\frac{3}{5}$	ddM1 A1
	So $AX:XN=4:1$	A1 (7)
ALT	$\left(\overrightarrow{AX}=\overrightarrow{AB}+\overrightarrow{BN}+\lambda\left(\pm\left(\text{their }\overrightarrow{AN}\right)\right)\right)=\mathbf{b}-\mathbf{a}-\frac{1}{4}\mathbf{b}+\lambda\left(\pm"-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)$	B1ft
	$\left(\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(\pm\left(\text{their }\overrightarrow{MB}\right)\right)=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm"-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)$	M1
	Equating components gives $-\frac{1}{2}-\frac{1}{2}\mu=-1+\lambda$	M1
	and $\mu=\frac{3}{4}-\frac{3}{4}\lambda$	M1
	$\mu=\frac{3}{4}-\frac{3}{4}\left(\frac{1}{2}-\frac{1}{2}\mu\right)\Rightarrow\lambda=\frac{1}{5}$ *or* $\mu=\frac{3}{5}$	ddM1 A1
	So $AX:XN=4:1$	A1 (7)

General principles for marking part b

B1 ft states a valid vector, with a parameter, that could be used in a solution to find the ratio required.

M1 states a second valid vector path, with a second distinct parameter that could be used to find the ratio required. **This must be a distinct route, different to that used already, not travelling along the same line.**

M1 This mark is obtained by correctly equating their components for **a. They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ**

M1 This mark is obtained by correctly equating their components for **b. They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ**

ddM1 For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ or their parameters.

Dependent on both previous method marks.

A1 correct value for one of their parameters

A1 correct ratio

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Part	Mark	Additional Guidance
(a) (i)	M1	For a correct vector path stated, a fully correct answer implies this mark.
	A1	For $-\mathbf{a}+\frac{3}{4}\mathbf{b}$
(ii)	B1 (A1 ePen)	For $-\mathbf{b}+\frac{1}{2}\mathbf{a}$
(b)	B1 ft	For $\overrightarrow{AX}=\lambda\pm$ their $\overrightarrow{AN}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX}=\right)\lambda\left(\pm\left("-\mathbf{a}+\frac{3}{4}\mathbf{b}"\right)\right)$ This may be seen embedded in working.
	M1	For e.g. $\overrightarrow{AX}=\overrightarrow{AM}+\mu\pm$ their $\overrightarrow{MB}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX}=\right)-\frac{1}{2}\mathbf{a}+\mu\left(\pm\left("-\frac{1}{2}\mathbf{a}+\mathbf{b}"\right)\right)$ This may be seen embedded in working. **This must be a distinct route, different to that used already, not travelling along the same line.**
	M1	For $-\lambda=-\frac{1}{2}-\frac{1}{2}\mu$ This mark is obtained by correctly equating their components for **a**, the example given is when their equations are set up using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	M1	For $\frac{3}{4}\lambda=\mu$ This mark is obtained by correctly equating their components for **b**, the example given is when their equations are set up using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	ddM1	For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ or their parameters. Dependent on both previous method marks.
	A1	There are a number of different correct answers here, depending on how they've set up their equations: For $\lambda=\frac{4}{5}$ or $\mu=\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ For $\lambda=-\frac{4}{5}$ or $\mu=\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\overrightarrow{MB}$ For $\lambda=\frac{4}{5}$ or $\mu=-\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\overrightarrow{AN}$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(-\overrightarrow{MB}\right)$ For $\lambda=-\frac{4}{5}$ or $\mu=-\frac{3}{5}$ using $\overrightarrow{AX}=\lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX}=\overrightarrow{AM}+\mu\left(-\overrightarrow{MB}\right)$
	A1	For $AX:XN=4:1$

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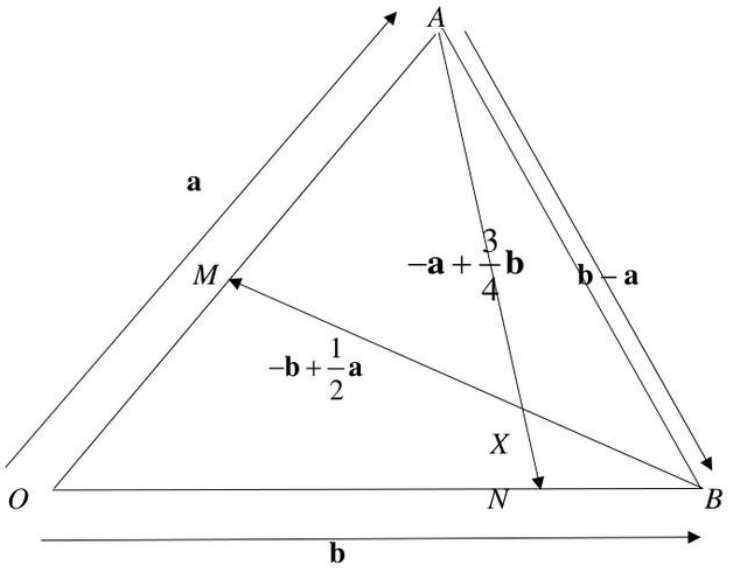
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(b) ALT	**B1ft**	For $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(\pm\text{their } \overrightarrow{AN}\right)$ written in terms of **a** and **b** $\left(\overrightarrow{AX}\right) = \mathbf{b} - \mathbf{a} - \frac{1}{4}\mathbf{b} + \lambda\left(\pm"-\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$ This may be seen embedded in working.
	M1	For e.g. $\overrightarrow{AX} = \overrightarrow{AM} + \mu \pm \text{their } \overrightarrow{MB}$ written in terms of vectors **a** and **b** $\left(\overrightarrow{AX} =\right) -\frac{1}{2}\mathbf{a} + \mu\left(\pm"-\frac{1}{2}\mathbf{a} + \mathbf{b}"\right)$ This may be seen embedded in working. **This must be a distinct route, different to that used already, not travelling along the same line.**
	M1	For $-\frac{1}{2} - \frac{1}{2}\mu = -1 + \lambda$ This mark is obtained by correctly equating their components for **a**, the example given is when their equations are set up using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	M1	For $\mu = \frac{3}{4} - \frac{3}{4}\lambda$ This mark is obtained by correctly equating their components for **b**, the example given is when their equations are set up using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ **They must have two relevant vectors, along two distinct paths and two distinct parameters. These do not need to be labelled λ and μ.**
	ddM1	For an attempt to solve their simultaneous equations. There can be errors, but it must be clear they are solving simultaneous equations to arrive at a value for λ or μ Dependent on both previous method marks.
	A1	There are a number of different correct answers here, depending on how they've set up their equations: For $\lambda = \frac{1}{5}$ or $\mu = \frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ For $\lambda = -\frac{1}{5}$ or $\mu = \frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\overrightarrow{MB}$ For $\lambda = \frac{1}{5}$ or $\mu = -\frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\overrightarrow{AN}$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\left(-\overrightarrow{MB}\right)$ For $\lambda = -\frac{1}{5}$ or $\mu = -\frac{3}{5}$ using $\overrightarrow{AX} = \overrightarrow{AB} + \overrightarrow{BN} + \lambda\left(-\overrightarrow{AN}\right)$ and $\overrightarrow{AX} = \overrightarrow{AM} + \mu\left(-\overrightarrow{MB}\right)$
	A1	For $AX : XN = 4 : 1$

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Type MS · Where lines cross: two routes and comparing coefficients

Answer 2 of 20 · Question 11 part b · 4 marks

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11(a)	(i) $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$	B1
	(ii) $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC} = \mathbf{a} + \frac{1}{2}\left(-\mathbf{a} + \frac{1}{4}\mathbf{b}\right) = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$	M1A1
	(iii) $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD} = -\mathbf{b} + \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$	M1A1 (5)
(b)	$\overrightarrow{BE} = k\overrightarrow{BD} = k\left(\frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}\right)$	M1
	$\overrightarrow{BE} = -\mathbf{b} + \overrightarrow{OE} = -\mathbf{b} + \lambda\mathbf{a}$	M1
	$\frac{7}{8}k = 1 \Rightarrow k = \frac{8}{7}$	M1
	$\frac{k}{2} = \lambda \Rightarrow \lambda = \frac{4}{7}$	A1 (4)
(c)	$\triangle OAC = \frac{1}{4}\triangle OAB$	M1
	$\triangle OEB = "\frac{4}{7}"\triangle OAB$	A1 ft
	$\frac{\triangle OAC}{\triangle OEB} = \frac{1}{4} \times "\frac{7}{4}" = \frac{7}{16}$	M1
	$\mu = \frac{7}{16}$	A1 (4)
		[13]

Part	Mark	Notes
(a)(i)	B1	For the correct vector $\overrightarrow{AC} = -\mathbf{a} + \frac{1}{4}\mathbf{b}$
(ii)	M1	For the correct vector statement $\overrightarrow{OD} = \overrightarrow{OA} + \frac{1}{2}\overrightarrow{AC}$ or $\overrightarrow{OD} = \overrightarrow{OC} + \frac{1}{2}\overrightarrow{CA}$
	A1	For the correct **simplified** vector $\overrightarrow{OD} = \frac{1}{2}\mathbf{a} + \frac{1}{8}\mathbf{b}$
(iii)	M1	For the correct vector statement $\overrightarrow{BD} = \overrightarrow{BO} + \overrightarrow{OD}$ or $\overrightarrow{BD} = \overrightarrow{BC} + \overrightarrow{CD}$
	A1	For the correct **simplified** vector $\overrightarrow{BD} = \frac{1}{2}\mathbf{a} - \frac{7}{8}\mathbf{b}$

Type MS · Where lines cross: two routes and comparing coefficients

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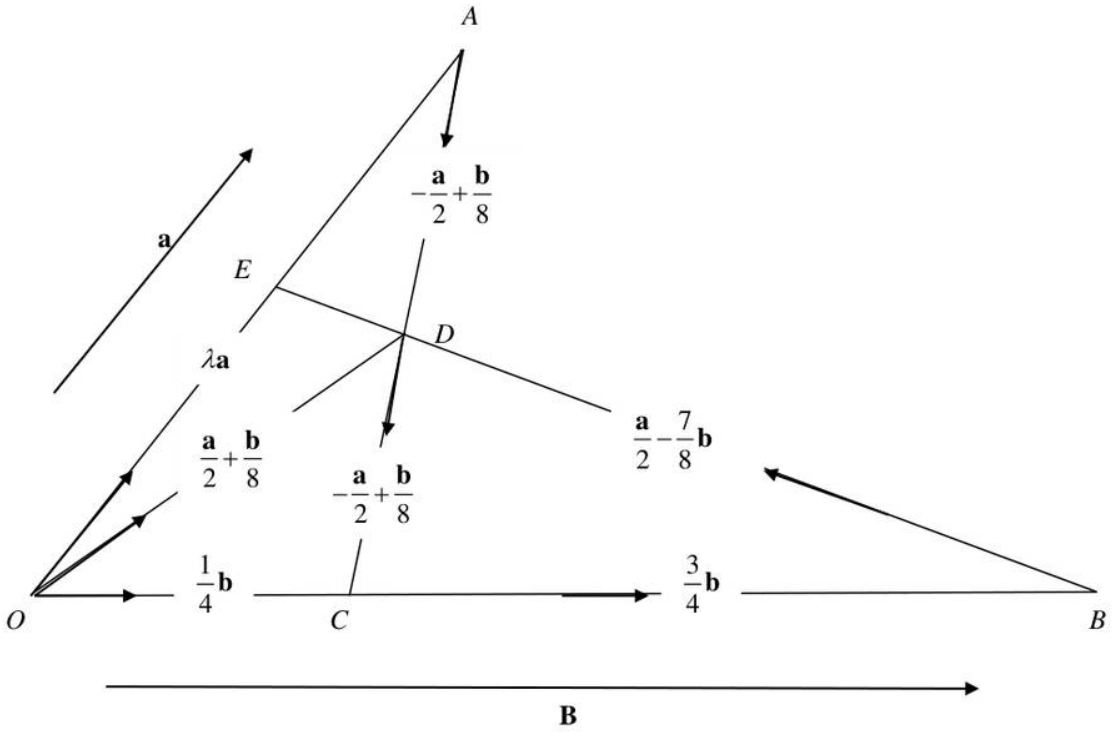
In parts (b) and (c) you must follow through the vectors they have found in part (a)		
(b)	M1	States or implies two paths that will lead to a solution. For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}$ and $\overrightarrow{BE}=\overrightarrow{BO}+\overrightarrow{OE}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\overrightarrow{OD}+\overrightarrow{DE}$ **This is a B mark in Epen**
	M1	For writing their paths as vectors in terms of **a**, **b** λ **and another constant** (e.g. k or μ) For example: using triangle OEB • $\overrightarrow{BE}=k\overrightarrow{BD}=k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ and $\overrightarrow{BE}=-\mathbf{b}+\lambda\mathbf{a}$ **or** using triangle OED • $\overrightarrow{OE}=\lambda\mathbf{a}$ and $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ OR $\overrightarrow{OE}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$
	dM1	Equates coefficients of their **two** expressions and attempts to find the value of λ: In triangle OEB • $k\frac{1}{2}\mathbf{a}-k\frac{7}{8}\mathbf{b}=\lambda\mathbf{a}-\mathbf{b}$ $\Rightarrow k\frac{7}{8}=1\Rightarrow k=\frac{8}{7}$ and $\frac{k}{2}=\lambda\Rightarrow\lambda=...$ or in triangle OED, there are two possible expressions for $\overrightarrow{DE}$ • $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(-\mathbf{b}+\lambda\mathbf{a}\right)$ $\Rightarrow k=\frac{1}{8}$ and $\lambda=\frac{1}{2}+\frac{1}{8}\lambda$ $\Rightarrow\lambda=...$ Or $\lambda\mathbf{a}=\frac{1}{2}\mathbf{a}+\frac{1}{8}\mathbf{b}+k\left(\frac{1}{2}\mathbf{a}-\frac{7}{8}\mathbf{b}\right)$ $\Rightarrow\frac{7}{8}k=\frac{1}{8}\Rightarrow k=\frac{1}{7}$ and $\lambda=\frac{1}{2}+\frac{1}{7}k\Rightarrow\lambda...$
	A1	For the correct value of λ $\lambda=\frac{4}{7}$
(c)	M1	For stating that $\triangle OAC=\frac{1}{4}\triangle OAB$ or $4\triangle OAC=\triangle OAB$ **1**
	M1	For stating $\triangle OEB="\lambda"\triangle OAB\Rightarrow\triangle OEB="\frac{4}{7}"\triangle OAB$ **2** **This is an A mark in Epen**
	M1	For dividing **1** by **2** $=\frac{\triangle OAC}{\triangle OEB}=\frac{1}{4}\times"\frac{7}{4}"=...$
	A1	For $\mu=\frac{7}{16}$

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USEFUL SKETCH



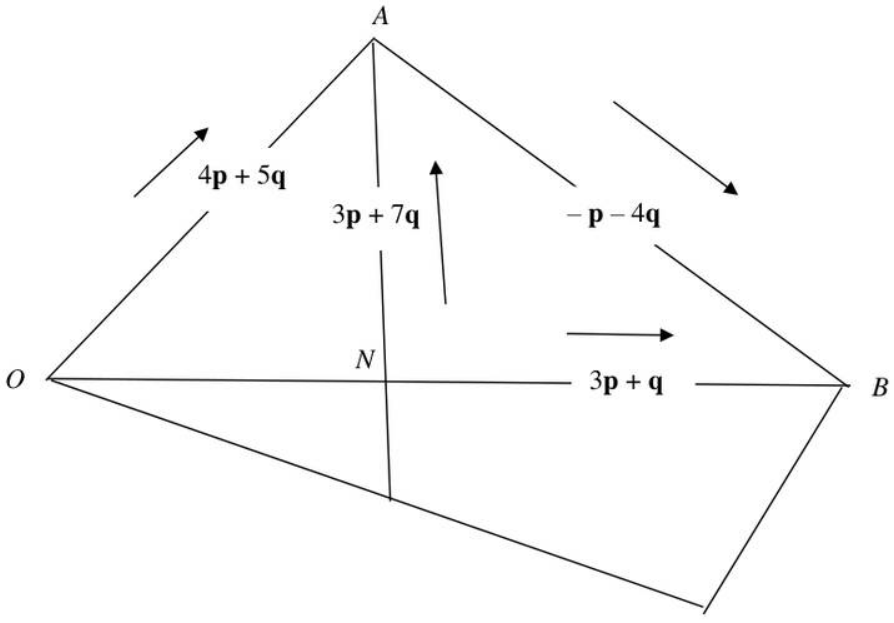
Type MS · Where lines cross: two routes and comparing coefficients

Answer 3 of 20 · Question 11 part b · 6 marks

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11(a)	$\overrightarrow{OM} = \frac{1}{2}\overrightarrow{OC} = \mathbf{p} - 2\mathbf{q}$ $\overrightarrow{MA} = -\overrightarrow{OM} + \overrightarrow{OA}$ $\overrightarrow{MA} = -(\mathbf{p} - 2\mathbf{q}) + (4\mathbf{p} + 5\mathbf{q}) = 3\mathbf{p} + 7\mathbf{q}$	B1 M1 A1 [3]
(b)	$\overrightarrow{MN} = \lambda(3\mathbf{p} + 7\mathbf{q})$ $\overrightarrow{MN} = -(\mathbf{p} - 2\mathbf{q}) + \mu(3\mathbf{p} + \mathbf{q})$ $\lambda(3\mathbf{p} + 7\mathbf{q}) = -(\mathbf{p} - 2\mathbf{q}) + \mu(3\mathbf{p} + \mathbf{q})$ $\Rightarrow 3\lambda\mathbf{p} + 7\lambda\mathbf{q} = (-1 + 3\mu)\mathbf{p} + (2 + \mu)\mathbf{q}$ $3\lambda = -1 + 3\mu$ $7\lambda = 2 + \mu$ $\Rightarrow 18\lambda = 7 \Rightarrow \lambda = \frac{7}{18}$ $MN : NA = 7 : 11$	M1 M1 M1 dM1 A1 A1 [6]
		Total 9 marks

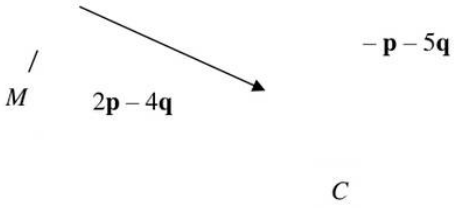
Useful sketch



Type MS · Where lines cross: two routes and comparing coefficients

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Part	Mark	Notes
(a)	B1	For finding either vector $\overrightarrow{OM} = \mathbf{p} - 2\mathbf{q}$ or $\overrightarrow{MO} = -\mathbf{p} + 2\mathbf{q}$ This may be embedded in their working for $\overrightarrow{MA}$
	M1	For the correct vector statement $\overrightarrow{MA} = -\overrightarrow{OM} + \overrightarrow{OA}$ or $\overrightarrow{MA} = -\frac{1}{2}\overrightarrow{OC} + \overrightarrow{OA}$
	A1	For the correct simplified vector
(b)	**General principle of marking part (b)** • First two M marks are for two vector statements for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ that will allow them to be equated. Note that they can use any path that involves either of these two vectors. • The third M mark is for equating coefficients and forming a pair of simultaneous equations. • The final M mark is for solving their simultaneous equations.	
	M1	For one vector for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ involving a constant One example is: $\overrightarrow{AN} = K\overrightarrow{AM} = K(-3\mathbf{p} - 7\mathbf{q})$
	M1	For a second vector for $\overrightarrow{MN}$ or $\overrightarrow{AN}$ following a different path involving a different constant. One example is: $\overrightarrow{AN} = \overrightarrow{AO} + L\overrightarrow{OB} = -4\mathbf{p} - 5\mathbf{q} + L(3\mathbf{p} + \mathbf{q})$
	M1	For equating the two vectors and forming a pair of simultaneous linear equations, both of which must be in terms of λ and μ
	dM1	For an attempt to solve their linear equations. Allow up to one processing error. This mark is dependent on the previous M mark.
	A1	For the value of $\lambda = \frac{7}{18}$ μ is not required but the value is $\frac{13}{18}$
	A1	For the correct ratio $MN : NA = 7 : 11$

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11(a)	$\overrightarrow{AB} = -\mathbf{a}+\mathbf{b}$ $\overrightarrow{AC} = -\mathbf{a}+\frac{2}{3}\mathbf{b}$ $\overrightarrow{AM} = -\mathbf{a}+\frac{5}{6}\mathbf{b}$	B2
	$\overrightarrow{NB} = -\frac{1}{2}\overrightarrow{AC}+\overrightarrow{AB} = -\frac{1}{2}\left(-\mathbf{a}+\frac{2}{3}\mathbf{b}\right)+\left(-\mathbf{a}+\mathbf{b}\right) = \left[-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right]$ OR $\overrightarrow{NB} = -\frac{1}{2}\overrightarrow{AC}+\overrightarrow{AB} = \frac{1}{2}\left(-\mathbf{a}+\frac{2}{3}\mathbf{b}\right)+\frac{1}{3}\mathbf{b} = \left[-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right]$	M1
	Two statements for $\overrightarrow{OP}$ using two parameters $\overrightarrow{OP} = \overrightarrow{OA}+\lambda\overrightarrow{AM} = \mathbf{a}+\lambda\left(-\mathbf{a}+\frac{5}{6}\mathbf{b}\right) = \left[\mathbf{a}(1-\lambda)+\frac{5}{6}\lambda\mathbf{b}\right]$	M1
	$\overrightarrow{OP} = \overrightarrow{OB}-\mu\overrightarrow{NB} = \mathbf{b}-\mu\left(-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right) = \left[\mathbf{a}\left(\frac{\mu}{2}\right)+\mathbf{b}\left(1-\frac{2\mu}{3}\right)\right]$	M1
	Equates coefficients $\Rightarrow 1-\lambda = \frac{\mu}{2}$ $\frac{5}{6}\lambda = 1-\frac{2\mu}{3}$	M1
	Solves S. E. $\Rightarrow \lambda = \frac{2}{3}$ $\mu = \frac{2}{3}$	M1
	$\Rightarrow \overrightarrow{OP} = \mathbf{a}+\frac{2}{3}\left(-\mathbf{a}+\frac{5}{6}\mathbf{b}\right) = \frac{1}{3}\mathbf{a}+\frac{5}{9}\mathbf{b}$	M1A1 [9]
(b)	Any two from $\overrightarrow{CP} = \frac{1}{3}\mathbf{a}-\frac{1}{9}\mathbf{b}$ $\overrightarrow{PQ} = \frac{1}{6}\mathbf{a}-\frac{1}{18}\mathbf{b}$ $\overrightarrow{CQ} = \frac{1}{2}\mathbf{a}-\frac{1}{6}\mathbf{b}$	B2
	$\overrightarrow{CQ} = \frac{3}{2}\overrightarrow{CP}$ OR $\overrightarrow{PQ} = 2\overrightarrow{CP}$ OR $\overrightarrow{CQ} = 3\overrightarrow{PQ}$	M1
	with a conclusion	A1 [4]
		Total 13 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 4 of 20 · Question 11 part a · 9 marks

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Part	Mark	Notes
(a)	B1	For at least one from $\overrightarrow{AB} = -\mathbf{a}+\mathbf{b}$ $\overrightarrow{AC} = -\mathbf{a}+\frac{2}{3}\mathbf{b}$ $\overrightarrow{AM} = -\mathbf{a}+\frac{5}{6}\mathbf{b}$
	B1	All three correct.
	M1	For a correct vector statement for $\overrightarrow{NB}$
	There are several paths to $\overrightarrow{OP}$. Those below are examples. For the award of each of the two M marks, there must be two different paths using two distinct parameters.	
	M1	For either vector statement for $\overrightarrow{OP}$ using a parameter e.g. $\overrightarrow{OP} = \overrightarrow{OA}+\lambda\overrightarrow{AM} = \mathbf{a}+\lambda\left(-\mathbf{a}+\frac{5}{6}\mathbf{b}\right) = \left[\mathbf{a}(1-\lambda)+\frac{5}{6}\lambda\mathbf{b}\right]$
	M1	For a second vector statements for $\overrightarrow{OP}$ using a different parameter e.g. $\overrightarrow{OP} = \overrightarrow{OB}-\mu\overrightarrow{NB} = \mathbf{b}-\mu\left(-\frac{\mathbf{a}}{2}+\frac{2\mathbf{b}}{3}\right) = \left[\mathbf{a}\left(\frac{\mu}{2}\right)+\mathbf{b}\left(1-\frac{2\mu}{3}\right)\right]$
	M1	For equating coefficients of **a** and **b**
	dM1	For solving the simultaneous equations This mark is dependent on the previous 3 M marks $\left[\lambda = \frac{2}{3} \quad \text{OR} \quad \mu = \frac{2}{3}\right]$
	M1	For substituting their value of λ or μ into their $\overrightarrow{OP} = \overrightarrow{OB}-\mu\overrightarrow{NB}$ or $\overrightarrow{OP} = \overrightarrow{OA}+\lambda\overrightarrow{AM}$
	A1	For the correct vector $\overrightarrow{OP} = \frac{1}{3}\mathbf{a}+\frac{5}{9}\mathbf{b}$
	SC – If $\overrightarrow{CQ}$ is assumed to be a straight line and is used to find used $\overrightarrow{OP}$, award all the marks as above but withhold the final A mark.	
(b)	B2	For any two vectors from $\overrightarrow{CP} = \frac{1}{3}\mathbf{a}-\frac{1}{9}\mathbf{b}$ or $\overrightarrow{PQ} = \frac{1}{6}\mathbf{a}-\frac{1}{18}\mathbf{b}$ or $\overrightarrow{CQ} = \frac{1}{2}\mathbf{a}-\frac{1}{6}\mathbf{b}$ (B1 for one of the above)
	M1	For $\overrightarrow{CQ} = \frac{3}{2}\overrightarrow{CP}$ oe or $\overrightarrow{PQ} = 2\overrightarrow{CP}$ oe or $\overrightarrow{CQ} = 3\overrightarrow{PQ}$
	A1cso	For a correct conclusion with supporting working

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Type MS · Where lines cross: two routes and comparing coefficients

Answer 5 of 20 · Question 8 part b · 5 marks

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8(a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DC} + \overrightarrow{CB} = 5\mathbf{a} - 2\mathbf{b} - 3\mathbf{a} = 2\mathbf{a} - 2\mathbf{b}$	B1 [1]
(b)	$\overrightarrow{BX} = k\overrightarrow{BD} = k(3\mathbf{a} + 2\mathbf{b})$ $\overrightarrow{BX} = \overrightarrow{BA} + \lambda\overrightarrow{AC} = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow k(3\mathbf{a} + 2\mathbf{b}) = -2\mathbf{a} + 2\mathbf{b} + \lambda(5\mathbf{a} - 2\mathbf{b})$ $\Rightarrow 3k\mathbf{a} + 2k\mathbf{b} = (5\lambda - 2)\mathbf{a} + (2 - 2\lambda)\mathbf{b}$ $\Rightarrow 3k = 5\lambda - 2 \quad 2k = 2 - 2\lambda$ $\Rightarrow k = \frac{3}{8} \quad \left[\lambda = \frac{5}{8}\right]$	M1 M1 M1 M1A1 [5]
(c)	$\Delta CXD = \frac{5}{8}\Delta BCD$ $\Delta BCD = \frac{3}{8}ABCD$ $\frac{\Delta CXD}{ABCD} = \frac{\frac{5}{8}}{\frac{8}{3}} = \frac{15}{64}$ Ratio of area of triangle CXD : Trapezium $ABCD$ = 15 : 64	M1 M1 M1 A1 [4]
		Total 10 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 6 of 20 · Question 9 part d · 6 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part d in this mark scheme extract

9(a)	$\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} \Rightarrow \overrightarrow{OP} = 2\mathbf{a} + \frac{1}{2}(4\mathbf{b} - 2\mathbf{a}) = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(c)	$\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}(\overrightarrow{OP})$ $\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a} + 2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$	M1 B1 A1 [3]
(d)	$\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$ $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$ $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \left(\frac{8}{5}\right)$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \frac{12}{20}, \left(\frac{3}{5}\right)$ $OR : RB = 3 : 2$	M1 M1 M1 M1A1 A1 [6]
	Total	**13 marks**

Part	Mark	Scheme
(a)	M1	For the correct vector statement $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB}$
	A1	$\overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$
(b)	M1	For a correct vector statement e.g $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}$
	A1	For the correct simplified vector $\overrightarrow{OP} = 2\mathbf{a} + (2\mathbf{b} - \mathbf{a}) = \mathbf{a} + 2\mathbf{b}$
(c)	M1	For a correct vector statement e.g., $\overrightarrow{AQ} = \overrightarrow{AO} + \overrightarrow{OQ}$ or $\overrightarrow{AQ} = \overrightarrow{AP} + \overrightarrow{PQ}$
	B1	For the correct vector statement for $\overrightarrow{OQ}$ $\overrightarrow{OQ} = \frac{3}{4}\left(\overrightarrow{OP}\right)$ or $\overrightarrow{PQ} = -\frac{1}{4}\left(\overrightarrow{OP}\right)$
	A1	For the correct simplified vector

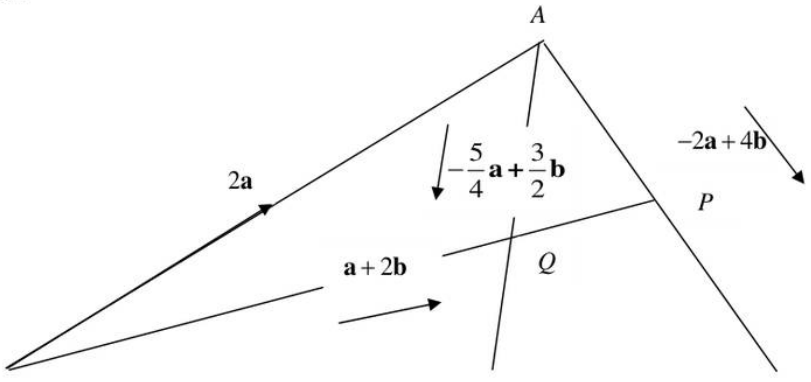
Type MS · Where lines cross: two routes and comparing coefficients

Answer 6 of 20 · Question 9 part d · 6 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part d in this mark scheme extract

		$\overrightarrow{AQ} = -2\mathbf{a} + \frac{3}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$ or $\overrightarrow{AQ} = \frac{1}{2}(-2\mathbf{a}+4\mathbf{b}) - \frac{1}{4}(\mathbf{a}+2\mathbf{b}) = -\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}$
(d)	In this part there is more than one route to finding the required ratio, but any route must involve $\overrightarrow{OR}$ or $\overrightarrow{RB}$ As the general principle: The first M mark is for one vector using a parameter. The second M mark is for a second vector using a different parameter. The third M mark is for equating coefficients and forming two equations in both parameters. The fourth M mark is for solving their equations. Check their working and do not allow an erroneous method. The following is using one path. Please trace their path on the sketch to check that it is valid. The path they use must involve the vector $\overrightarrow{OR}$ or $\overrightarrow{RB}$ eg., $\overrightarrow{AR} = \overrightarrow{AO} + \overrightarrow{OR}$ or $\overrightarrow{AR} = \overrightarrow{AB} + \overrightarrow{BR}$	
	M1	For the correct statement $\overrightarrow{OR} = \lambda 4\mathbf{b}$ or $\overrightarrow{OR} = \phi\mathbf{b}$
	M1	For the correct statement for $\overrightarrow{OR}$ $\overrightarrow{OR} = \overrightarrow{OA} + \mu\overrightarrow{AQ} = 2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right)$
	M1	For equating the two statements for $\overrightarrow{OR}$ and equating coefficients $2\mathbf{a} + \mu\left(-\frac{5}{4}\mathbf{a} + \frac{3}{2}\mathbf{b}\right) = \lambda 4\mathbf{b}$ $\mathbf{a}\left(2 - \frac{5\mu}{4}\right) + \frac{3\mu}{2}\mathbf{b} = \lambda 4\mathbf{b} \Rightarrow \frac{3\mu}{2} = 4\lambda$ and $2 - \frac{5\mu}{4} = 0$
	M1	For solving their two simultaneous equations **correctly** to find the value of λ (the parameter for OR) $2 - \frac{5\mu}{4} = 0 \Rightarrow \mu = \frac{8}{5}$ $\frac{3}{2} \times \frac{8}{5} = 4\lambda \Rightarrow \lambda = \ldots$
	A1	For the correct value of λ $\lambda = \frac{12}{20} = \left(\frac{3}{5}\right)$
	A1	For the correct ratio $OR : RB = 3 : 2$

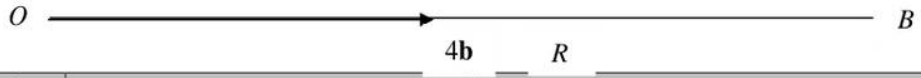
Useful Sketch



Type MS · Where lines cross: two routes and comparing coefficients

Answer 6 of 20 · Question 9 part d · 6 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part d in this mark scheme extract



Question	Scheme	Mark

Type MS · Where lines cross: two routes and comparing coefficients

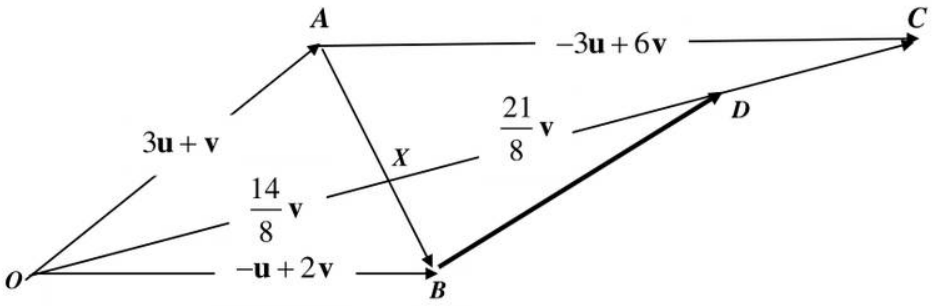
Answer 7 of 20 · Question 8 part b · 4 marks

2025 JUN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

8 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB}, = -(3\mathbf{u}+\mathbf{v}) + (2\mathbf{v}-\mathbf{u}) = -4\mathbf{u}+\mathbf{v}$	M1,A1 (2)
(b)	$\overrightarrow{AC} = \lambda(2\mathbf{v}-\mathbf{u})$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \overrightarrow{AC} = (3\mathbf{u}+\mathbf{v}) + \lambda(2\mathbf{v}-\mathbf{u}) = \mu\mathbf{v}$	M1
	$\mathbf{u}: 0 = 3-\lambda \qquad \mathbf{v}: \mu = 1+2\lambda$	M1
	$\mu = 7 \qquad [\lambda = 3]$	A1 (4)
(c)	$\overrightarrow{OX} = \overrightarrow{OA} + \alpha\overrightarrow{AB} = 3\mathbf{u}+\mathbf{v}+\alpha(-4\mathbf{u}+\mathbf{v})$ or $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AC} + \beta\overrightarrow{CO} = 3\mathbf{u}+\mathbf{v}+3(2\mathbf{v}-\mathbf{u})+\beta(-7\mathbf{v})$	M1
	$\overrightarrow{OX} = \frac{7}{4}\mathbf{v}$	A1
	$\overrightarrow{OD} = \frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD} = \frac{3}{2}\overrightarrow{OX}$	M1
	$\overrightarrow{OD} = \frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD} = \frac{21}{8}\mathbf{v}$	M1
	$\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{35}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}-2\mathbf{v}+\frac{7}{4}\mathbf{v}+\frac{21}{8}\mathbf{v}$ or $\overrightarrow{BD} = \mathbf{u}+\frac{19}{8}\mathbf{v}$	A1 (5)
		Total 11 marks

USEFUL SKETCH

$$\overrightarrow{OD} = \frac{35}{8}\mathbf{v} \quad \text{and} \quad \overrightarrow{AX} = -3\mathbf{u}+\frac{3}{4}\mathbf{v}$$



Type MS · Where lines cross: two routes and comparing coefficients

Answer 7 of 20 · Question 8 part b · 4 marks

2025 JUN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	M1	Use of $\overrightarrow{AB}=\overrightarrow{AO}+\overrightarrow{OB}$
	A1	$-4\mathbf{u}+\mathbf{v}$
(b)	M1	For use of $\overrightarrow{AC}$ is parallel to $\overrightarrow{OB}$ to form an expression for $\overrightarrow{AC}$
	M1	For use of their expression for $\overrightarrow{AC}$ to form an expression for $\overrightarrow{OC}$
	M1	For equating for **u** and for **v**
	A1	For $\mu=7$
(c)		**General principles for marking part (c)** For example: You may see the vector path for $\overrightarrow{BD}$ first - score the 4th M mark • Candidates will need to find vector $\overrightarrow{OX}$ [M1A1] • Candidates will need to use the ratio of bases to find $\overrightarrow{OD}=\frac{5}{2}\overrightarrow{OX}$ or $\overrightarrow{XD}=\frac{3}{2}\overrightarrow{OX}$ **leading** to establish $\overrightarrow{OD}=\frac{35}{8}\mathbf{v}$ or $\overrightarrow{XD}=\frac{21}{8}\mathbf{v}$ [M1] • Check any path they choose on the diagram below. For example, $\overrightarrow{BD}=\overrightarrow{BO}+\overrightarrow{OX}+\overrightarrow{XD}$ or [M1] $\overrightarrow{BD}=\overrightarrow{BO}+\overrightarrow{OA}+\overrightarrow{AX}+\overrightarrow{XD}$ If a candidate chooses to take a longer path and need to find for example, $\overrightarrow{AX}$ they will need to find a vector. There is no separate mark for this. • For the correct vector
	M1	For an expression for $\overrightarrow{OX}$ including an unknown constant
	A1	For $\overrightarrow{OX}=\frac{7}{4}\mathbf{v}$
	M1	For recognising that triangle OBX and triangle OBD share a base OB and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{OD}$ together with an attempt to find $\overrightarrow{OD}$ OR For recognising that triangle OBX and triangle BXD share a base BX and using this to obtain a ratio for $\overrightarrow{OX}$ to $\overrightarrow{XD}$ together with an attempt to find $\overrightarrow{XD}$
	M1	For method to find $\overrightarrow{BD}$ Award this for any valid path given
	A1	$\overrightarrow{BD}=\mathbf{u}+\frac{19}{8}\mathbf{v}$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 7 of 20 · Question 8 part b · 4 marks

2025 JUN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 8 of 20 · Question 10 part c · 7 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part c in this mark scheme extract

10 (a)	$\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -(2\mathbf{a} - \mathbf{b}) + 3\mathbf{a} + \mathbf{b} = \mathbf{a} + 2\mathbf{b}$	M1A1 [2]
(b)	$\overrightarrow{OC} = \overrightarrow{OB} + \overrightarrow{BC} = 3\mathbf{a} + \mathbf{b} - \mathbf{a} + 3\mathbf{b} = 2\mathbf{a} + 4\mathbf{b} = 2(\mathbf{a} + 2\mathbf{b})$ Conclusion required; same direction and $\overrightarrow{OC}$ is a multiple of $\overrightarrow{AB}$ therefore $\overrightarrow{OC}$ is parallel to $\overrightarrow{AB}$.	M1 A1 [2]
(c)	$\overrightarrow{AC} = \overrightarrow{AB} + \overrightarrow{BC} = \mathbf{a} + 2\mathbf{b} + (-\mathbf{a} + 3\mathbf{b}) = 5\mathbf{b}$ $\overrightarrow{AX} = \mu\overrightarrow{AC} = \mu 5\mathbf{b}$ $\overrightarrow{AX} = \overrightarrow{AO} + \overrightarrow{OX} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow \mu 5\mathbf{b} = -(2\mathbf{a} - \mathbf{b}) + \lambda(3\mathbf{a} + \mathbf{b})$ $\Rightarrow -2 + 3\lambda = 0 \Rightarrow \lambda = \frac{2}{3}$ $\Rightarrow 5\mu = 1 + \lambda \Rightarrow \mu = \frac{1}{3}$ $\Rightarrow AX : XC = 1 : 2$	B1 B1 M1 M1 M1 A1 A1 [7]
		Total 11 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 8 of 20 · Question 10 part c · 7 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 9 of 20 · Question 10 part b · 7 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

10 (a)	$\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$	M1
	$\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$	M1
	$\overrightarrow{DC} = \frac{9}{2}(\mathbf{p} + \mathbf{q}) = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$	A1 (3)
(b)	$\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$	M1
	$= 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$	A1
	$\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$	M1
	$= 5\mathbf{p} + 3\mu\mathbf{q}$	A1
	$3 + \frac{9}{2}\lambda = 5$	M1
	$\lambda = \frac{4}{9}$ $\left(\frac{9}{2}\lambda = 3\mu\right)$ or $\mu = \frac{2}{3}$	A1
	$\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$	A1 (7)
(c)	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	M1
	$\overrightarrow{OG} = 3\mathbf{p} + 2(\mathbf{p} + \mathbf{q}) + 5\beta\mathbf{p} = 5\mathbf{p} + 5\beta\mathbf{p} + 2\mathbf{q}$ ft their $\overrightarrow{OF}$	M1
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$ $\left(\beta = -\frac{1}{3}\right)$	M1
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	A1 (4)
	ALT (c)	
	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	{M1}
	$\overrightarrow{FG} = -5\mathbf{p} - 2\mathbf{q} + \alpha(5\mathbf{p} + 3\mathbf{q})$ ft their $-\left(\overrightarrow{OF}\right)$	{M1}
	As FG is parallel to AO there can be no $\mathbf{q}$ component in $\overrightarrow{FG}$	
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$	{M1}
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	{A1} (4)
		Total 14 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 9 of 20 · Question 10 part b · 7 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For finding the vector $\overrightarrow{OB}$: $\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$ **This must be correct.**
	M1	For the correct vector statement for $\overrightarrow{DC}$: $\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$ This mark can be implied by a correct (unsimplified) vector.
	A1	For the correct simplified $\overrightarrow{DC}$ in terms of a single **p** and a single **q** only $\overrightarrow{DC} = -3\mathbf{p} + \frac{3}{2}(5\mathbf{p} + 3\mathbf{q}) = \frac{9}{2}(\mathbf{p} + \mathbf{q})$ accept $\overrightarrow{DC} = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$ oe
(b)	M1	For stating $\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$ (or any other variable in place of λ)
	A1	For the vector $\overrightarrow{OF} = 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$
	M1	For stating $\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$ (or any other variable in place of μ)
	A1	For the vector $\overrightarrow{OF} = 5\mathbf{p} + 3\mu\mathbf{q}$
	M1	Collects like terms correctly, equates coefficients correctly and attempts to solve their two equations in λ and μ They must achieve a value for μ or λ for this mark $3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q}) = 5\mathbf{p} + 3\mu\mathbf{q} \Rightarrow \mathbf{p}\left(3 + \frac{9}{2}\lambda\right) + \frac{9}{2}\lambda\mathbf{q} = 5\mathbf{p} + 3\mu\mathbf{q}$ $3 + \frac{9}{2}\lambda = 5$ and $\frac{9}{2}\lambda = 3\mu$ $\Rightarrow \mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct $\mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct vector $\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$
		Part (b) can be solved in different ways. These are the general principles of marking this part question. **M1** – for any correct vector statement leading to $\overrightarrow{OF}$ which introduces a parameter. **A1** – for one correct vector $\overrightarrow{OF}$ **M1** - For any correct second vector statement for $\overrightarrow{OF}$ which can be used with the first vector and introduces a second parameter. **A1** – For a second correct vector $\overrightarrow{OF}$ **M1A1A1** – as main scheme above
(c)	M1	For one vector for $\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 9 of 20 · Question 10 part b · 7 marks

2021 JUN · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

	M1	For a second vector for $\overrightarrow{OG} = \left(\overrightarrow{OF} + \overrightarrow{FG}\right) = 3\mathbf{p} + 2(\mathbf{p}+\mathbf{q}) + 5\beta\mathbf{p} = \left(5\mathbf{p} + 5\beta\mathbf{p} + 2\mathbf{q}\right)$ ft their $\overrightarrow{OF}$ Accept simplified or unsimplified.
	M1	Collects like terms correctly, equates coefficients correctly to find the value of α $\alpha(5\mathbf{p}+3\mathbf{q}) = 5\mathbf{p}+5\beta\mathbf{p}+2\mathbf{q} \Rightarrow 5\mathbf{p}\alpha + 3\mathbf{q}\alpha = (5+5\beta)\mathbf{p} + 2\mathbf{q}$ $\Rightarrow 3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$
	ALT	
	M1	For the vector for $\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$
	M1	For the vector $\overrightarrow{FG} = \left(\overrightarrow{FO} + \overrightarrow{OG}\right) = -5\mathbf{p} - 2\mathbf{q} + \alpha\ 5\mathbf{p} + 3\mathbf{q}$ **Note:** ft their $-\left(\overrightarrow{OF}\right)$ from part (b)
	M1	As FG is parallel to OA there is no $\mathbf{q}$ component in $\overrightarrow{FG}$ $\Rightarrow 3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 9 of 20 · Question 10 part b · 7 marks

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Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 10 of 20 · Question 10 part c · 4 marks

2021 JUN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Number		
10 (a)	$\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$	M1
	$\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$	M1
	$\overrightarrow{DC} = \frac{9}{2}(\mathbf{p} + \mathbf{q}) = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$	A1 (3)
(b)	$\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$	M1
	$= 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$	A1
	$\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$	M1
	$= 5\mathbf{p} + 3\mu\mathbf{q}$	A1
	$3 + \frac{9}{2}\lambda = 5$	M1
	$\lambda = \frac{4}{9}$ $\left(\frac{9}{2}\lambda = 3\mu\right)$ or $\mu = \frac{2}{3}$	A1
	$\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$	A1 (7)
(c)	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	M1
	$\overrightarrow{OG} = 3\mathbf{p} + 2(\mathbf{p} + \mathbf{q}) + 5\beta\mathbf{p} = 5\mathbf{p} + 5\beta\mathbf{p} + 2\mathbf{q}$ ft their $\overrightarrow{OF}$	M1
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$ $\left(\beta = -\frac{1}{3}\right)$	M1
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	A1 (4)
	ALT (c)	
	$\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$	{M1}
	$\overrightarrow{FG} = -5\mathbf{p} - 2\mathbf{q} + \alpha(5\mathbf{p} + 3\mathbf{q})$ ft their $-\left(\overrightarrow{OF}\right)$	{M1}
	As FG is parallel to AO there can be no $\mathbf{q}$ component in $\overrightarrow{FG}$	
	$3\alpha = 2 \Rightarrow \alpha = \frac{2}{3}$	{M1}
	$\overrightarrow{OG} = \frac{10}{3}\mathbf{p} + 2\mathbf{q}$	{A1} (4)
		Total 14 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 10 of 20 · Question 10 part c · 4 marks

2021 JUN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For finding the vector $\overrightarrow{OB}$: $\left(\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB}\right) = 5\mathbf{p} + 3\mathbf{q}$ **This must be correct.**
	M1	For the correct vector statement for $\overrightarrow{DC}$: $\overrightarrow{DC} = \left(\overrightarrow{DO} + \overrightarrow{OC}\right) = \overrightarrow{DO} + \frac{3}{2}\overrightarrow{OB}$ This mark can be implied by a correct (unsimplified) vector.
	A1	For the correct simplified $\overrightarrow{DC}$ in terms of a single **p** and a single **q** only $\overrightarrow{DC} = -3\mathbf{p} + \frac{3}{2}(5\mathbf{p} + 3\mathbf{q}) = \frac{9}{2}(\mathbf{p} + \mathbf{q})$ accept $\overrightarrow{DC} = \frac{9\mathbf{p}}{2} + \frac{9\mathbf{q}}{2}$ oe
(b)	M1	For stating $\overrightarrow{OF} = \overrightarrow{OD} + \lambda\overrightarrow{DC}$ (or any other variable in place of λ)
	A1	For the vector $\overrightarrow{OF} = 3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q})$
	M1	For stating $\overrightarrow{OF} = \overrightarrow{OA} + \mu\overrightarrow{AB}$ (or any other variable in place of μ)
	A1	For the vector $\overrightarrow{OF} = 5\mathbf{p} + 3\mu\mathbf{q}$
	M1	Collects like terms correctly, equates coefficients correctly and attempts to solve their two equations in λ and μ They must achieve a value for μ or λ for this mark $3\mathbf{p} + \frac{9}{2}\lambda(\mathbf{p} + \mathbf{q}) = 5\mathbf{p} + 3\mu\mathbf{q} \Rightarrow \mathbf{p}\left(3 + \frac{9}{2}\lambda\right) + \frac{9}{2}\lambda\mathbf{q} = 5\mathbf{p} + 3\mu\mathbf{q}$ $3 + \frac{9}{2}\lambda = 5$ and $\frac{9}{2}\lambda = 3\mu$ $\Rightarrow \mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct $\mu = \frac{2}{3}$ or $\lambda = \frac{4}{9}$
	A1	For the correct vector $\overrightarrow{OF} = 5\mathbf{p} + 2\mathbf{q}$
		Part (b) can be solved in different ways. These are the general principles of marking this part question. **M1** – for any correct vector statement leading to $\overrightarrow{OF}$ which introduces a parameter. **A1** – for one correct vector $\overrightarrow{OF}$ **M1** - For any correct second vector statement for $\overrightarrow{OF}$ which can be used with the first vector and introduces a second parameter. **A1** – For a second correct vector $\overrightarrow{OF}$ **M1A1A1** – as main scheme above
(c)	M1	For one vector for $\overrightarrow{OG} = \alpha(5\mathbf{p} + 3\mathbf{q})$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 10 of 20 · Question 10 part c · 4 marks

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	M1	For a second vector for $\overrightarrow{OG}=\left(\overrightarrow{OF}+\overrightarrow{FG}\right)=3\mathbf{p}+2(\mathbf{p}+\mathbf{q})+5\beta\mathbf{p}=\left(5\mathbf{p}+5\beta\mathbf{p}+2\mathbf{q}\right)$ ft their $\overrightarrow{OF}$ Accept simplified or unsimplified.
	M1	Collects like terms correctly, equates coefficients correctly to find the value of α $\alpha(5\mathbf{p}+3\mathbf{q})=5\mathbf{p}+5\beta\mathbf{p}+2\mathbf{q}\Rightarrow 5\mathbf{p}\alpha+3\mathbf{q}\alpha=(5+5\beta)\mathbf{p}+2\mathbf{q}$ $\Rightarrow 3\alpha=2\Rightarrow\alpha=\frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG}=\frac{10}{3}\mathbf{p}+2\mathbf{q}$
	ALT	
	M1	For the vector for $\overrightarrow{OG}=\alpha(5\mathbf{p}+3\mathbf{q})$
	M1	For the vector $\overrightarrow{FG}=\left(\overrightarrow{FO}+\overrightarrow{OG}\right)=-5\mathbf{p}-2\mathbf{q}+\alpha\ 5\mathbf{p}+3\mathbf{q}$ **Note:** ft their $-\left(\overrightarrow{OF}\right)$ from part (b)
	M1	As FG is parallel to OA there is no $\mathbf{q}$ component in $\overrightarrow{FG}\Rightarrow 3\alpha=2\Rightarrow\alpha=\frac{2}{3}$
	A1	For the correct vector $\overrightarrow{OG}=\frac{10}{3}\mathbf{p}+2\mathbf{q}$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 10 of 20 · Question 10 part c · 4 marks

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Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 11 of 20 · Question 9 part b · 6 marks

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9 (a)	$\overrightarrow{AB} = \overrightarrow{AD} + \overrightarrow{DB} = 2\mathbf{a}+\mathbf{b} - -4\mathbf{a}-\mathbf{b} = 6\mathbf{a}+2\mathbf{b}$ or $2\ 3\mathbf{a}+\mathbf{b}$ oe	B1
	$\overrightarrow{DC} = \overrightarrow{DB} + \overrightarrow{BC} = 4\mathbf{a}+\mathbf{b}+\frac{1}{3}\mathbf{b} = \left(4\mathbf{a}+\frac{4}{3}\mathbf{b}\right)$ or $\frac{4}{3}(3\mathbf{a}+\mathbf{b})$	B1
	$\overrightarrow{AB} = \frac{3}{2}\overrightarrow{DC}$ so $\overrightarrow{AB}$ is a multiple of $\overrightarrow{DC}$	M1
	Conclusion: Therefore, DC parallel to AB)	A1
		[4]
(b)	$\overrightarrow{AC} = 2\mathbf{a}+\mathbf{b}+'\left(4\mathbf{a}+\frac{4}{3}\mathbf{b}\right)' = \left[6\mathbf{a}+\frac{7}{3}\mathbf{b}\right]$	M1
	$\overrightarrow{AY} = \lambda\text{“}\overrightarrow{AC}\text{”}$ $(=\lambda\text{”}(6\mathbf{a}+\frac{7}{3}\mathbf{b})\text{”})$	M1
	$\overrightarrow{AY} = \overrightarrow{AD} + \mu\overrightarrow{DB}$ $(= 2\mathbf{a}+\mathbf{b}+\mu(4\mathbf{a}+\mathbf{b}))$ or $\overrightarrow{AY} = \text{“}\overrightarrow{AB}\text{”} + \alpha\overrightarrow{BD}$ $(=\text{“}6\mathbf{a}+2\mathbf{b}\text{”}+\alpha(-4\mathbf{a}-\mathbf{b}))$	M1
	$2+4\mu = 6\lambda$ or $6-4\alpha = 6\lambda$ $1+\mu = \frac{7}{3}\lambda$ $1+\mu = \frac{7}{3}\lambda$	M1
	$\lambda = \frac{3}{5}$ or $\mu = \frac{2}{5}$ or $\alpha = \frac{3}{5}$	A1
	$\overrightarrow{AY} = \frac{18}{5}\mathbf{a}+\frac{7}{5}\mathbf{b}$ or $\overrightarrow{AY} = \frac{1}{5}(8\mathbf{a}+7\mathbf{b})$	A1cao
		[6]
		Total 10 marks

Part	Mark	Additional Guidance
(a)	B1	For stating a valid vector for $\overrightarrow{AB} = 6\mathbf{a}+2\mathbf{b}$ [accept simplified or unsimplified] **NB: This is an M mark in Epen**
	B1	For stating a valid vector for $\overrightarrow{DC} = \frac{4}{3}(3\mathbf{a}+\mathbf{b})$ [accept simplified or unsimplified] **NB: This is an A mark in Epen**
	M1	For comparing their two vectors and establishing that $\overrightarrow{DC}$ and $\overrightarrow{AB}$ are multiples of each other .i.e. $\overrightarrow{AB} = \frac{3}{2}\overrightarrow{DC}$ oe **NB** This statement must be correct! **Incorrect directions.** Award this mark if they are comparing for example; $\overrightarrow{CD}$ and $\overrightarrow{AB}$ or $\overrightarrow{CD}$ and $\overrightarrow{BA}$ and ft their incorrect direction for the M mark.

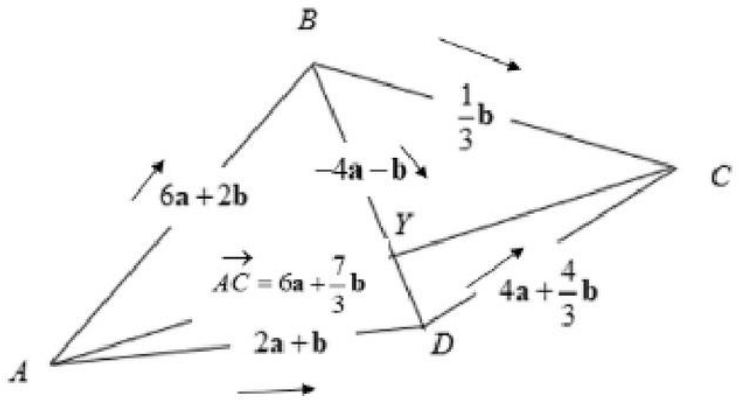
Type MS · Where lines cross: two routes and comparing coefficients

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	A1ft	For a conclusion that the vectors are therefore parallel. Allow for example $\overrightarrow{AB} = -6\mathbf{a} - 2\mathbf{b}$ and $\overrightarrow{DC} = \frac{4}{3}(3\mathbf{a} + \mathbf{b})$ are therefore parallel Accept a very brief conclusion even if it is just the word 'shown' or 'QED' or even '#' or a tick.
(b)	Part (b) can be done in a number of different ways – use the following as general principles to mark, referring also to the scheme and additional guidance for the example given, to determine when each mark should be awarded	
	M1	For any simplified or un-simplified vector along AC $\overrightarrow{AC} = 2\mathbf{a} + \mathbf{b} + '\left(4\mathbf{a} + \frac{4}{3}\mathbf{b}\right)' = \left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$ For this mark allow $\overrightarrow{CA} = -6\mathbf{a} - \frac{7}{3}\mathbf{b}$ as it is possible to use this to find $\overrightarrow{AY}$ This mark can be implied by a correct vector for $\overrightarrow{AY}$ e.g. $\overrightarrow{AY} = \lambda\left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$
	M1	Uses their vector for $\overrightarrow{AC}$ to write a vector for $\overrightarrow{AY}$ by introducing a parameter e.g. $\overrightarrow{AY} = \lambda\left[6\mathbf{a} + \frac{7}{3}\mathbf{b}\right]$
	M1	States a second valid vector path for $\overrightarrow{AY}$ introducing a second parameter that can be used with the first vector path [to solve an equation to find their parameters]. Note: this mark is awarded only for stating the vector path, but it must be valid – i.e. able to be used to find the parameters. This second parameter cannot be 1 – their first parameter.
	dddM1	Equating components and reaching a value for any of their parameters. Allow arithmetical slips in processing their simultaneous equations. This is dependent on all previous method marks.
	A1	Any one correct value for one of their two parameters.
	A1	For $\overrightarrow{AY} = \frac{18}{5}\mathbf{a} + \frac{7}{5}\mathbf{b}$

USEFUL SKETCH



Type MS · Where lines cross: two routes and comparing coefficients

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Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 12 of 20 · Question 10 part b · 5 marks

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10(a)	(i) $\overrightarrow{AB} = -\overrightarrow{OA} + \overrightarrow{OB} \Rightarrow \overrightarrow{AB} = -2\mathbf{a} + 4\mathbf{b}$	B1
	(ii) $\overrightarrow{MY} = \overrightarrow{MA} + \frac{3}{4}\left(\overrightarrow{AB}\right) = \mathbf{a} + \frac{3}{4}(-2\mathbf{a} + 4\mathbf{b}) = -\frac{\mathbf{a}}{2} + 3\mathbf{b}$	M1A1 [4]
(b)	$\overrightarrow{OX} = \mu\overrightarrow{OB} = \mu 4\mathbf{b}$	M1
	$\overrightarrow{OX} = \overrightarrow{OM} + \overrightarrow{MX} = \overrightarrow{OM} + \lambda\overrightarrow{MY} = \mathbf{a} + \lambda\left(-\frac{\mathbf{a}}{2} + 3\mathbf{b}\right) = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\lambda\mathbf{b}$	M1
	$\Rightarrow \mu 4\mathbf{b} = \mathbf{a}\left(1 - \frac{\lambda}{2}\right) + 3\mathbf{b}$	dM1
	$\Rightarrow 1 - \frac{\lambda}{2} = 0 \Rightarrow \lambda = 2$	ddM1
	$\Rightarrow 4\mu = 3\lambda \Rightarrow \mu = \frac{6}{4} = \frac{3}{2}$	
	$OB : OX = 2:3$ oe	A1 [5]
	ALT – working with alternative vector within triangle *OMX*	
	$\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OB} + \mu\overrightarrow{OB} = -\boldsymbol{a} + 4\boldsymbol{b} + \mu 4\boldsymbol{b}$	[M1
	$\overrightarrow{MX} = \lambda\left(-\frac{\boldsymbol{a}}{2} + 3\boldsymbol{b}\right)$	M1
	$\Rightarrow -\boldsymbol{a} = -\frac{\lambda\boldsymbol{a}}{2} \Rightarrow \lambda = 2$	dM1
	$\Rightarrow 4\boldsymbol{b} + \mu 4\boldsymbol{b} = 3\lambda\mathbf{b} \Rightarrow \mu = \frac{1}{2}$	ddM1
	$OB : OX = 2:3$ oe	A1]
(c)	$\frac{\Delta YBX}{\Delta ABX} = \frac{1}{4}$	M1
	$\frac{\Delta ABX}{\Delta OAX} = \frac{1}{3}$	
	$\Rightarrow \frac{\Delta YBX}{\Delta OAX} = \frac{1}{4} \times \frac{1}{3} = \frac{1}{12} \Rightarrow \Delta YBX : \Delta OAX = 1:12$	M1A1 [3]
	ALT – working with relative areas of triangles	
	Area $\Delta YBX = a$ Area $\Delta ABX = 4a$ Area $\Delta OMX = 6a$ Area $\Delta AYX = 3a$ Area $\Delta OAY = 6a$ Area $\Delta OYB = 2a$	[M1
	Area $\Delta OAX = 12a$	M1
	$\Delta YBX : \Delta OAX = 1:12$	A1]
		Total 11 marks

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Part	Mark	Notes
(a)	B1	For the correct simplified vector $\overrightarrow{AB}$
	M1	For the correct vector statement for $\overrightarrow{MY}$
	A1	For the correct simplified vector for $\overrightarrow{MY}$
(b)	M1	For the statement $\overrightarrow{OX} = \mu 4\mathbf{b}$ Note: this is a B mark on epen.
	M1	For the correct vector for $\overrightarrow{OX}$ (ft their MY)
	dM1	For equating both vectors for $\overrightarrow{OX}$ and for comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5
	ALT – working with alternative vector within triangle OMX	
	M1	For a vector statement which includes $\mu 4\boldsymbol{b}$ (for OX or BX) (ft their MY) Note: this is a B mark on epen.
	M1	For a correct second vector equation for the same vector (ft their MY)
	dM1	For equating both vectors and comparing coefficients of **a** and **b** Dep on M1M1
	ddM1	For finding a value for their parameter for μ Note: there is no mark for only finding λ, they must find μ Dep on M1M1M1
	A1	For the correct ratio $OB:OX = 2:3$ Allow equivalent ratios e.g. 4 : 6, 1 : 1.5 Condone $OX:OB = 3:2$ if clearly stated, but not just 3:2
(c)	M1	For either the relationship between the of areas of triangles BYX and ABX **or** the relationship between the areas of triangles ABX and OAX
	M1	For finding the relationship between the of areas of triangles BYX and OAX
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).
	ALT – working with relative areas	
	M1	For assigning a value to one triangle area and writing a second area in terms of this. Note: This could also follow from working with area of a triangle $= \frac{1}{2}ab\sin C$

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		e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	M1	For finding the relationship between the of areas of triangles BYX and OAX Note: This could also follow from working with area of a triangle $= \frac{1}{2} ab \sin C$ e.g. $\Delta YBX = \frac{1}{2} yz \sin B$ and $\Delta ABX = \frac{1}{2} y(4z) \sin B$
	A1	For the correct ratio [1:12] Allow equivalent ratios. Note: do not penalise answer given as a fraction i.e. $\frac{1}{12}$ if already penalised in (b).

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Type MS · Where lines cross: two routes and comparing coefficients

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10 (a) (i)	$\mathbf{a}+\mathbf{c}$	B1
(ii)	$\frac{1}{2}(\mathbf{c}-\mathbf{a})$	B1 (2)
(b)	$\overrightarrow{OX} = OA + AM + \lambda MN$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a})$	M1 A1
	$\mu(\mathbf{a}+\mathbf{c})$	B1
	$\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{1}{2}\mathbf{c}-\frac{1}{2}\mathbf{a}\right)=\mu(\mathbf{a}+\mathbf{c})$	M1
	$1-\frac{1}{2}\lambda=\mu \quad \frac{1}{2}+\frac{1}{2}\lambda=\mu$	M1
	$1-\frac{1}{2}\lambda=\frac{1}{2}+\frac{1}{2}\lambda$	M1
	$\lambda=\frac{1}{2} \quad \mu=\frac{3}{4}$	A1 A1 (8)
(c)	Triangle $XBN=\frac{1}{8}$ of $\frac{1}{2}$ the parallelogram Quadrilateral $OXNC=\frac{7}{8}$ of $\frac{1}{2}$ the parallelogram So Quadrilateral $OXNC=\frac{7}{16}$ of the parallelogram $\therefore$ 7 : 16	M1 M1 A1 (3) **[13]**

Part	Mark	Additional Guidance
(a)(i)	B1	For the correct vector $\mathbf{a}+\mathbf{c}$
(ii)	B1	For the correct vector $\frac{1}{2}(\mathbf{c}-\mathbf{a})$
(b)	M1	For the correct vector statement $\overrightarrow{OX} = \overrightarrow{OA} + \overrightarrow{AM} + \lambda\overrightarrow{MN}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ or $\overrightarrow{OX}=\mathbf{a}+\frac{1}{2}\mathbf{c}+\lambda\left(\frac{\mathbf{c}}{2}-\frac{\mathbf{a}}{2}\right)$
	B1ft	For $\overrightarrow{OX}=\mu(\mathbf{a}+\mathbf{c})$ ft their $\overrightarrow{OB}=$'$\mathbf{a}+\mathbf{c}$'
	M1	For equating their two vector statements for $\overrightarrow{OX}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}+\mathbf{c})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $\mathbf{a}+\frac{1}{2}\mathbf{c}+\frac{\lambda}{2}(\mathbf{c}-\mathbf{a})=\mu(\mathbf{a}-\mathbf{c})\Rightarrow\mathbf{a}\left(1-\frac{\lambda}{2}\right)+\mathbf{c}\left(\frac{1}{2}+\frac{\lambda}{2}\right)=\mu\mathbf{a}+\mu\mathbf{c}$ $\Rightarrow\mu=1-\frac{\lambda}{2},\ \mu=\frac{1}{2}+\frac{\lambda}{2}$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $1-\frac{\lambda}{2}=\frac{1}{2}+\frac{\lambda}{2}\Rightarrow\lambda=\ldots\Rightarrow\mu=\ldots$ or $1-\mu=\mu-\frac{1}{2}\Rightarrow\mu=\ldots\Rightarrow\lambda=\ldots$

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	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
	ALT	
	M1	For the correct vector statement $\overrightarrow{MX} = \overrightarrow{MO} + \overrightarrow{OX}$
	A1	For the correct vector (need not be simplified) $\overrightarrow{MX} = -\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c})$
	B1ft	$\overrightarrow{MX} = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$ ft their $\overrightarrow{MN} = '\frac{1}{2}(\mathbf{c}-\mathbf{a})'$
	M1	For equating the two vector statements for $\overrightarrow{MX}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a})$
	M1	For equating coefficients of $\mathbf{a}$ and $\mathbf{c}$ $-\frac{\mathbf{c}}{2} - \mathbf{a} + \mu(\mathbf{a}+\mathbf{c}) = \frac{\lambda}{2}(\mathbf{c}-\mathbf{a}) \Rightarrow \mathbf{c}\left(-\frac{1}{2}+\mu\right) + \mathbf{a}(\mu-1) = \mathbf{c}\frac{\lambda}{2} - \mathbf{a}\frac{\lambda}{2}$ $\Rightarrow \frac{\lambda}{2} = \mu - \frac{1}{2}$ and $-\frac{\lambda}{2} = \mu - 1$
	M1	For attempting to solve their two simultaneous equations in terms of λ and μ. $\mu - \frac{1}{2} = -(\mu - 1) \Rightarrow \mu = \left(\frac{3}{4}\right)$ $\frac{\lambda}{2} = 1 - '\frac{3}{4}' \Rightarrow \lambda = \left(\frac{1}{2}\right)$
	A1	For **either** $\lambda = \frac{1}{2}$ **or** $\mu = \frac{3}{4}$
	A1	For **both** $\lambda = \frac{1}{2}$ **and** $\mu = \frac{3}{4}$
(c)	M1	For area of $\Delta XBN = \frac{1}{8}\Delta OBC$ so $\frac{1}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ $\left(\Delta OBC = \frac{1}{2} \times OB \times BC \times \sin \angle XBN\right.$ $\left.\Delta XBN = \frac{1}{2} \times \frac{1}{4} OB \times \frac{1}{2} BC \times \sin \angle XBN = \frac{1}{8}\Delta OBC\right)$
	M1	Therefore Quadrilateral $OXNC = \frac{7}{8}$ of $\frac{1}{2}$ of the area of parallelogram $OABC$ ft their fraction from the first M mark provided it is $< \frac{1}{2}$
	A1	Quadrilateral $OXNC = \frac{7}{16}$ of the area of parallelogram $OABC$ so ratio is 7:16

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Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 14 of 20 · Question 10 part b · 6 marks

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10 a (i)	$\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB} = \mathbf{a} + \frac{3}{4}(-\mathbf{a}+\mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA} = \mathbf{b} + \frac{1}{4}(\mathbf{a}-\mathbf{b})$	M1
	$\overrightarrow{OP} = \frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$	A1
(ii)	$\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP} = -\frac{1}{2}\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$	M1
	$\overrightarrow{MN} = -\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$	A1 (4)
b	$\overrightarrow{OC} = \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)
ALT	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$	B1
	$\overrightarrow{AN} = \left(\overrightarrow{AO} + \overrightarrow{ON}\right) = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right) = -\frac{7}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or $\overrightarrow{AN} = \overrightarrow{AP} + \overrightarrow{PN} =$	M1
	$\overrightarrow{AC} = \mu\overrightarrow{AN} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$	M1 (A1 on ePen)
	$= -\frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$	A1 (M1 on ePen)
	$\therefore \lambda = \frac{3}{8}\mu$ and $0 = 1 - \frac{7}{8}\mu \Rightarrow \mu = \frac{8}{7}$ $\therefore \lambda = \frac{3}{7}$	M1 (A1 on ePen)
	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$	A1 (6)

General principles for marking part (b) – if in any doubt about allocating marks – send to review

B1 Writes a valid vector with a parameter in terms of **a** or **b** which leads to finding $\overrightarrow{OC}$

M1 M1 A1 Writes a second valid vector with a different parameter, in terms of **a** and **b**, following a distinct different route, which leads to finding $\overrightarrow{OC}$

M1 Compares components with two different parameters and arrives at a value for μ or λ

A1 correct vector

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c	(Area triangle) $OAP = \frac{3}{4}$ (Area triangle) OAB	B1
	(Area triangle) $OMN = \frac{1}{4}$ (Area triangle) $OAP = \frac{3}{16}$ (Area triangle) OAB	B1
	(Area quadrilateral) $AMNP =$ $\frac{3}{4}$ (Area triangle) $OAB - \frac{3}{16}$ (Area triangle) OAB	M1
	$= \frac{9}{16}$ (Area triangle) OAB $\quad k = \frac{9}{16}$	A1 (4)
ALT	$\frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4}$	B1
	$\left(\frac{\text{Area triangle } OMN}{\text{Area triangle } OAP} = \frac{1}{4}\right) \Rightarrow \frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} = \frac{3}{4}$	B1
	$\frac{(\text{Area quadrilateral})MNAP}{(\text{Area triangle})OAP} \times \frac{(\text{Area triangle})OAP}{(\text{Area triangle})OAB} = \frac{3}{4} \times \frac{3}{4}$	M1
	$k = \frac{9}{16}$	A1 (4)
		Total 14 marks

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Part	Marks	Note
(a) (i)	M1	For stating **or** using $\overrightarrow{OP} = \overrightarrow{OA} + \frac{3}{4}\overrightarrow{AB}$ or for $\mathbf{a} + \frac{3}{4}(-\mathbf{a} + \mathbf{b})$ or $\overrightarrow{OB} + \frac{1}{4}\overrightarrow{BA}$ or $\mathbf{b} + \frac{1}{4}(\mathbf{a} - \mathbf{b})$ (can be implied by correct vector)
	A1	For $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$ or valid alternative form such as $\frac{\mathbf{a}+3\mathbf{b}}{4}$ or $\frac{1}{4}(\mathbf{a}+3\mathbf{b})$
(ii)	M1	For stating **or** using $\overrightarrow{MN} = \overrightarrow{MO} + \frac{1}{2}\overrightarrow{OP}$ or $\frac{1}{2}\overrightarrow{AO} + \frac{1}{2}\overrightarrow{OP}$ or $-\frac{1}{2}\mathbf{a} + \frac{1}{2}\left(\text{their } \overrightarrow{OP}\right)$
	A1	$-\frac{3}{8}\mathbf{a} + \frac{3}{8}\mathbf{b}$ or valid alternative form such as $\frac{-3\mathbf{a}+3\mathbf{b}}{8}$ or $\frac{1}{8}(-3\mathbf{a}+3\mathbf{b})$ (can be implied by correct vector)
(b)	B1	$\overrightarrow{OC} = \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	**Using** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\left(\text{their } \overrightarrow{AN}\right)$ in terms **a** and **b** $\overrightarrow{OC} = \overrightarrow{OA} + \mu\overrightarrow{AN} = \mathbf{a} + \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ Simplification not required
	A1 (M1 on ePen)	$= \mathbf{a} - \frac{7}{8}\mu\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ *or* $\left(1 - \frac{7}{8}\mu\right)\mathbf{a} + \frac{3}{8}\mu\mathbf{b}$ either of these forms, ready for comparing coefficients
	M1 (A1 on ePen)	For correctly equating their components with two different parameters and attempting to solve, reaching values for μ or λ We need to see two equations here, leading to a value for one of the parameters.
	A1	$\overrightarrow{OC} = \frac{3}{7}\mathbf{b}$
ALT	B1	$\overrightarrow{AC} = -\mathbf{a} + \lambda\mathbf{b}$ or any equivalent statement involving a parameter, in terms of **a** and **b**.
	M1	A fully correct method to find $\overrightarrow{AN}$ using their $\overrightarrow{OP}$ $\overrightarrow{AN} = -\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)$
	M1 (A1 on ePen)	$\overrightarrow{AC} = \mu\left(-\mathbf{a} + \frac{1}{2}\left("\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}"\right)\right)$ using their $\overrightarrow{AN}$
	A1 (M1 on ePen)	Correct vector for $\overrightarrow{AC}$
	M1 (A1 on ePen) A1	Marks allocated as main scheme
(c) BOTH SCHEMES	B1	Correct statement
	B1	Correct statement
	M1	Uses their statements to carry out a relevant calculation
	A1	Correct value for k

Type MS · Where lines cross: two routes and comparing coefficients

Answer 14 of 20 · Question 10 part b · 6 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

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Type MS · Where lines cross: two routes and comparing coefficients

Answer 15 of 20 · Question 10 part b · 8 marks

2024 JUN · Paper 02 · extract 1 of 4 · find part b in this mark scheme extract

10(a)	$\overrightarrow{BC} = -\overrightarrow{AB} + \overrightarrow{AC} = -(3\mathbf{a}+4\mathbf{b}) + 7\mathbf{a}+9\mathbf{b} = 4\mathbf{a}+5\mathbf{b}$	B1
	$\overrightarrow{DC} = -\overrightarrow{DA} + \overrightarrow{DC} = -(4\mathbf{a}+5\mathbf{b}) + 7\mathbf{a}+9\mathbf{b} = 3\mathbf{a}+4\mathbf{b}$	B1
	$AB \parallel DC$ and $BC \parallel AD$ hence the quadrilateral is a parallelogram	B1ft [3]
(b)	$\overrightarrow{CF} = -\frac{1}{3}(3\mathbf{a}+4\mathbf{b})$	B1
	$\overrightarrow{AF} = \overrightarrow{AC} + \overrightarrow{CF} = 7\mathbf{a}+9\mathbf{b} - -\frac{1}{3}(3\mathbf{a}+4\mathbf{b}) = \left[6\mathbf{a}+\frac{23}{3}\mathbf{b}\right]$	B1
	$\overrightarrow{AE} = \mu\overrightarrow{AF} = \mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right)$	M1
	$\overrightarrow{AE} = \overrightarrow{AB} + \overrightarrow{BE} = 3\mathbf{a}+4\mathbf{b}+\lambda(4\mathbf{a}+5\mathbf{b}) = \left[(3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}\right]$	M1
	$\mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = (3+4\lambda)\mathbf{a}+(4+5\lambda)\mathbf{b}$	
	$\Rightarrow 6\mu = 3+4\lambda \quad \frac{23}{3}\mu = 4+5\lambda$	ddM1
	$\Rightarrow \lambda = \frac{3}{2} \quad \left(\mu = \frac{3}{2}\right)$	M1A1
	$\overrightarrow{AE} = \frac{3}{2}\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = 9\mathbf{a}+\frac{23}{2}\mathbf{b}$	B1ft [8]
		Total 11 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 15 of 20 · Question 10 part b · 8 marks

2024 JUN · Paper 02 · extract 2 of 4 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct vector for $\overrightarrow{BC}$ or $\overrightarrow{CB}$
	B1	For the correct vector for $\overrightarrow{DC}$ or $\overrightarrow{CD}$
	B1ft	For the correct complete conclusion including stating that the quadrilateral is a parallelogram. Ft their vectors for reverse directions.
	ALT	
	B1	For the correct lengths of $\overrightarrow{AD}$ AND $\overrightarrow{BC}$ $\quad\quad\quad

Type MS · Where lines cross: two routes and comparing coefficients

Answer 15 of 20 · Question 10 part b · 8 marks

2024 JUN · Paper 02 · extract 3 of 4 · find part b in this mark scheme extract

	ALT	
	B1	For the correct vector for $\overrightarrow{CF}$ or $\overrightarrow{DF}$ $\left[\overrightarrow{CF} = -\frac{1}{3}(3\mathbf{a}+4\mathbf{b})\right]$ $\left[\overrightarrow{DF} = \frac{2}{3}(3\mathbf{a}+4\mathbf{b})\right]$
	B1	For the correct vector for $\overrightarrow{AF}$ using their $\overrightarrow{CF}$ or $\overrightarrow{DF}$ e.g., $\overrightarrow{AF} = \overrightarrow{AC}+\overrightarrow{CF} = 7\mathbf{a}+9\mathbf{b}-\frac{1}{3}(3\mathbf{a}+4\mathbf{b}) = \left[6\mathbf{a}+\frac{23}{3}\mathbf{b}\right]$
	M1	For **one** correct vector for $\overrightarrow{AE}$ using **their** $\overrightarrow{AF}$ and a parameter For example: $\overrightarrow{AE} = \mu\overrightarrow{AF} = \mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right)$
	M1	For a correct vector for $\overrightarrow{BE}$ using their $\overrightarrow{AE}$ $\overrightarrow{BE} = -3\mathbf{a}-4\mathbf{b}+\mu\left(6\mathbf{a}+\frac{23}{3}\mathbf{b}\right) = \left[(6\mu-3)\mathbf{a}+\left(\frac{23}{3}\mu-4\right)\mathbf{b}\right]$
	ddM1	For forming the correct ratio using the vector $\overrightarrow{BC}$ $\frac{(6\mu-3)}{\left(\frac{23}{3}\mu-4\right)} = \frac{4}{5}$
	dddM1	For solving the equation to find a value for μ
	A1	For $\mu = \frac{3}{2}$
	B1ft	For writing down a vector for $\overrightarrow{AE}$ using their $\overrightarrow{AF}$ and their λ in the required form For writing down a vector for $\overrightarrow{AE}$ using their $\overrightarrow{AF}$ and their λ in the required form, provided $1 < \lambda < 2$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 15 of 20 · Question 10 part b · 8 marks

2024 JUN · Paper 02 · extract 4 of 4 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 16 of 20 · Question 9 part c.i · 4 marks

2025 NOV · Paper 02 · extract 1 of 5 · find part c.i in this mark scheme extract

	Scheme	Marks
9(a)	$\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$ or $-(2\mathbf{a}+4\mathbf{b})+(7\mathbf{a}+7\mathbf{b})$	B1
	$= 5\mathbf{a}+3\mathbf{b}$	B1
	$\overrightarrow{BC}\left[= \frac{1}{2}(10\mathbf{a}+6\mathbf{b})=\right] \frac{1}{2}\overrightarrow{AD}$ oe eg $\frac{1}{2}(10\mathbf{a}+6\mathbf{b}) = 5\mathbf{a}+3\mathbf{b}$ eg $10\mathbf{a}+6\mathbf{b} = 2(5\mathbf{a}+3\mathbf{b})$	M1
	[therefore parallel] therefore trapezium or eg # or shown.	A1 [4]
(b)	$\left[\overrightarrow{BD} = -\overrightarrow{AB} + \overrightarrow{AD} = -(2\mathbf{a}+4\mathbf{b})+(10\mathbf{a}+6\mathbf{b}) =\right] 8\mathbf{a}+2\mathbf{b}$	B1
	$\left[\lvert BD\rvert =\right] \sqrt{8^2+2^2}\left[=\sqrt{68}=2\sqrt{17}\right]$	B1ft
	Unit vector is $\pm\frac{1}{\sqrt{68}}(8\mathbf{a}+2\mathbf{b})$ oe eg $\pm\frac{1}{\sqrt{17}}(4\mathbf{a}+\mathbf{b})$	M1A1 [4]

Type MS · Where lines cross: two routes and comparing coefficients

Answer 16 of 20 · Question 9 part c.i · 4 marks

2025 NOV · Paper 02 · extract 2 of 5 · find part c.i in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct unsimplified vector in **a** and **b** for $\overrightarrow{BC}$ or for $\left[\overrightarrow{BC} =\right] \overrightarrow{BA} + \overrightarrow{AC}$
	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BC}$
	M1	For $\overrightarrow{BC} = \frac{1}{2}\overrightarrow{AD}$ oe (note accept eg $2BC = AD$ ie without vector notation for this mark) Accept equivalences – see example MS.
	A1	For the correct conclusion, with all previous marks gained. Ignore work on other vectors. Do not accept eg $2BC = AD$ without vector notation for this mark Note B0 B1 M1 A0 is a potential score. $2BC = AD$ followed by eg $10\mathbf{a} + 6\mathbf{b} = 10\mathbf{a} + 6\mathbf{b}$ will get Aq
Watch for equivalences such as working out vector $\overrightarrow{CB}$ and stating $\overrightarrow{CB} = -\frac{1}{2}\overrightarrow{AD}$ Permit eg $\frac{\overrightarrow{AD}}{\overrightarrow{BC}} = 2$		
(b)	B1	For the correct simplified vector in **a** and **b** for $\overrightarrow{BD}$
	B1ft	For applying Pythagoras theorem fully and correctly to find the length BD Ft any vector believed to be their $\overrightarrow{BD}$. $8^2 + 2^2$ alone is insufficient.
	M1	For using their vector for $\overrightarrow{BD}$ with their length BD to attempt to form a unit vector
	A1	For the correct unit vector.

Type MS · Where lines cross: two routes and comparing coefficients

Answer 16 of 20 · Question 9 part c.i · 4 marks

2025 NOV · Paper 02 · extract 3 of 5 · find part c.i in this mark scheme extract

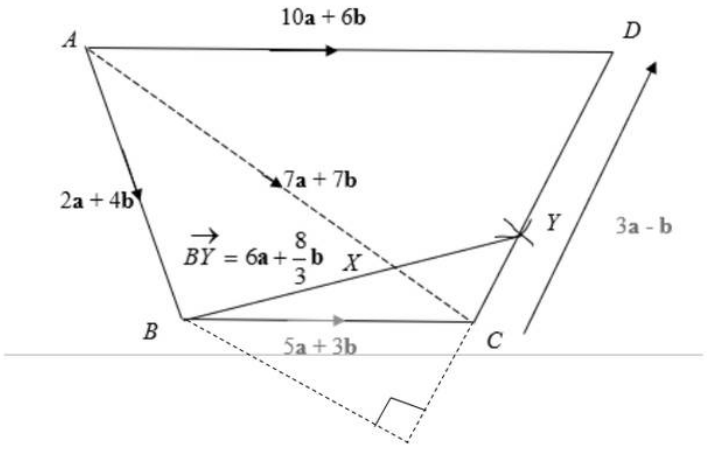
(c)(i)	$\left[\left[\overrightarrow{CY}=\right]\frac{1}{3}(3\mathbf{a}-\mathbf{b})=\mathbf{a}-\frac{1}{3}\mathbf{b}$ oe eg $\overrightarrow{YC}$ or $\left[\left[\overrightarrow{YD}=\right]\frac{2}{3}(3\mathbf{a}-\mathbf{b})=2\mathbf{a}-\frac{2}{3}\mathbf{b}$ oe eg $\overrightarrow{DY}$ Note: $\overrightarrow{BC}=5\mathbf{a}+4\mathbf{b}$ $\left[\overrightarrow{BY}=5\mathbf{a}+3\mathbf{b}+\mathbf{a}-\frac{\mathbf{b}}{3}=\right]6\mathbf{a}+\frac{8}{3}\mathbf{b}\Rightarrow$	M1
	$\left[\overrightarrow{BX}=\right]\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$	dM1
	$\left[\overrightarrow{BX}=\right]-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $\left[\overrightarrow{BX}=\right](5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})$ $\left[\lambda\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=-(2\mathbf{a}+4\mathbf{b})+\mu(7\mathbf{a}+7\mathbf{b})$ or $(5\mathbf{a}+3\mathbf{b})+\alpha(-7\mathbf{a}-7\mathbf{b})\right]$ $\left[\begin{array}{l}6\lambda=7\mu-2 \text{ and } \frac{8}{3}\lambda=7\mu-4 \text{ or}\\ \frac{8}{3}\lambda=-7\alpha+3 \text{ and } 6\lambda=-7\alpha+5\end{array}\right]$	ddM1
	$\left[\lambda=\frac{3}{5};\mu=\frac{4}{5};\alpha=\frac{1}{5}\Rightarrow\overrightarrow{BX}=\right]\frac{18}{5}\mathbf{a}+\frac{8}{5}\mathbf{b}\Rightarrow[BX:XY=]3:2$	A1 [4]
ALT	First mark as main scheme. $\left[\overrightarrow{XY}=\right]\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)$	M1
	$\left[\overrightarrow{XY}=\right]\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\left[\overrightarrow{XY}=\right]\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)$ $\left[\beta\left(6\mathbf{a}+\frac{8}{3}\mathbf{b}\right)=\gamma(7\mathbf{a}+7\mathbf{b})+\mathbf{a}-\frac{1}{3}\mathbf{b}$ or $\delta(-7\mathbf{a}-7\mathbf{b})+10\mathbf{a}+6\mathbf{b}-\left(2\mathbf{a}-\frac{2}{3}\mathbf{b}\right)\right]$	dM1
	$\left[\begin{array}{l}6\beta=7\gamma+1 \text{ and } \frac{8}{3}\beta=7\gamma-\frac{1}{3} \text{ or}\\ \frac{8}{3}\beta=-7\delta+6+\frac{2}{3} \text{ and } 6\beta=-7\delta+10-2\end{array}\right]$	ddM1
	$\left[\beta=\frac{2}{5};\gamma=\frac{1}{5};\delta=\frac{4}{5}\Rightarrow\overrightarrow{XY}=\right]\frac{12}{5}\mathbf{a}+\frac{16}{15}\mathbf{b}[BX:XY=]3:2$	A1 [4]

Type MS · Where lines cross: two routes and comparing coefficients

Answer 16 of 20 · Question 9 part c.i · 4 marks

2025 NOV · Paper 02 · extract 4 of 5 · find part c.i in this mark scheme extract

Part	Mark	Notes
(c)(i) Main and ALT	M1	Correct simplified vector in **a** and **b** for $\pm \overrightarrow{BY}$
	dM1	Correct vector in **a** and **b** with parameter using $\overrightarrow{CY}$ or $\overrightarrow{YC}$ for $\overrightarrow{BX}$ / $\overrightarrow{XB}$ or $\overrightarrow{XY}$ / $\overrightarrow{YX}$ **and** correct vector in **a** and **b** with a different parameter along a second distinct path using $\overrightarrow{AC}$ or $\overrightarrow{CA}$ for the same vector. Dependent on previous method mark.
	ddM1	For the correct simultaneous equations. Any one correct parameter or $\overrightarrow{BX} = \frac{18}{5}\mathbf{a} + \frac{8}{5}\mathbf{b}$ will imply this mark. Dependent on both previous method marks.
	A1	For the required ratio $BX : XY = 3 : 2$
Note, correct method can include the opposite signs in vectors multiplied by a parameter, as the parameter will just be the opposite sign when calculated and a fully correct solution is possible. Eg $\left[\overrightarrow{BX} = \right](5\mathbf{a}+3\mathbf{b}) + \alpha(7\mathbf{a}+7\mathbf{b})$ would just make $\alpha = -\frac{1}{5}$ Vectors questions notoriously have different methods. Correct answer, not from obviously incorrect working, full marks may be awarded **if method is shown.** (This is because the demand of this question says students must use a vector method). Min method must involve finding two correct different distinct relevant paths.		



Type MS · Where lines cross: two routes and comparing coefficients

Answer 16 of 20 · Question 9 part c.i · 4 marks

2025 NOV · Paper 02 · extract 5 of 5 · find part c.i in this mark scheme extract

(ii)	$\left[\frac{CXY}{BXC}=\frac{2}{3}\ \text{and}\ \frac{BAC}{BXC}=\frac{5}{1}\Rightarrow\right]\ \frac{CXY}{BAC}=\frac{2}{15}$ oe eg $\frac{BAC}{CXY}=\frac{15}{2}$	M1
Eg 1	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{CXY}{ABCD}=\frac{\frac{2}{15}}{3}\left[=\frac{2}{45}\right]$ oe eg $\frac{ABCD}{CXY}=\frac{3}{\frac{2}{15}}\left[=\frac{45}{2}\right]$	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
Eg 2	$\frac{CXY}{BCY}=\frac{2}{5}$ and $\left[\frac{BCY}{BCD}=\frac{1}{3}\right]\ \frac{BCY}{BCA}=\frac{1}{3}$ oe	M1
	$\frac{ABCD}{BAC}=\frac{3}{1}$ oe eg $\frac{BAC}{ABCD}=\frac{1}{3}$	M1
	$\frac{BCY}{ABCD}=\frac{\frac{1}{3}}{3}\left[=\frac{1}{9}\right]$ oe and $\frac{CXY}{ABCD}=\frac{\frac{2}{5}}{9}\left[=\frac{2}{45}\right]$ oe	M1
	$[\Rightarrow CXY:ABCD=]2:45$	A1 [4]
		Total 16 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 17 of 20 · Question 8 part d · 5 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

8 (a)	$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \mathbf{a} + \mathbf{b}$	B1 [1]
(b)	$\overrightarrow{OC} = 2\mathbf{b}$ $\overrightarrow{BC} = \overrightarrow{BO} + \overrightarrow{OC} = -(\mathbf{a} + \mathbf{b}) + 2\mathbf{b} = \mathbf{b} - \mathbf{a}$	B1 M1A1 [3]
(c)	$\overrightarrow{OM} = \overrightarrow{OB} + \overrightarrow{BM} = \mathbf{a} + \mathbf{b} + \frac{2}{3}(\mathbf{b} - \mathbf{a}) = \frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}$ or $\frac{1}{3}(\mathbf{a} + 5\mathbf{b})$	 M1A1ft [2]
(d)	$\overrightarrow{OY} = \mu\left(\frac{\mathbf{a}}{3} + \frac{5\mathbf{b}}{3}\right) = \frac{\mu\mathbf{a}}{3} + \frac{5\mu\mathbf{b}}{3}$ $\overrightarrow{OY} = \overrightarrow{OA} + \overrightarrow{AY} = \boldsymbol{a} + \lambda\boldsymbol{b}$ $\Rightarrow \frac{\mu}{3} = 1$ and $\frac{5\mu}{3} = \lambda$ Solves simultaneous equations by any method $\mu = 3,\ \lambda = 5$ $AB : BY = 1 : 4$ **ALT** $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ $\Rightarrow -1 + \frac{1}{3}\mu = 0$ and $\lambda = -1 + \frac{5}{3}\mu$ $\mu = 3, \lambda = 4$ $AB : BY = 1 : 4$	M1 M1 M1 M1 A1 [5] [M1 M1 M1 M1 A1]
(e)	$10 = \frac{1}{2}ab\sin 60^\circ \Rightarrow ab = \frac{40}{\sqrt{3}} \Rightarrow a = \frac{40}{b\sqrt{3}}$ $\text{Area} = \frac{1}{2}a \times 5b\sin 120^\circ = \frac{1}{2} \times \frac{40}{b\sqrt{3}} \times 5b\sin 120^\circ$ $\text{Area} = 50$ **ALT** $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2} \times h \times 5}{\frac{1}{2} \times h \times 1} = \frac{5}{1}$ $\text{Area } OAY = 5 \times \text{Area } OAB$ $\text{Area } OAB = 10$ $\text{Area} = 50$	M1A1 dM1 A1 [4] [M1 A1 dM1 A1]
		Total 15 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 17 of 20 · Question 8 part d · 5 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

Part	Mark	Notes
(a)	B1	For a correct expression for $\overrightarrow{OB}$ in terms of **a** and **b**
(b)	B1	For a correct expression for $\overrightarrow{OC}$
	M1	For a correct vector statement for $\overrightarrow{BC}$: $\overrightarrow{BC}=\overrightarrow{BO}+\overrightarrow{OC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$. Vector statement must be suitable for substitution to find $\overrightarrow{BC}$
	A1	For the correct simplified $\overrightarrow{BC}$ in terms of a single **a** and **b** only. $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ If answer $\overrightarrow{BC} = \mathbf{b} - \mathbf{a}$ seen without wrong working then award B1M1A1.
(c)	M1	For a correct vector statement for $\overrightarrow{OM}$: $\overrightarrow{OM}=\overrightarrow{OB}+\frac{2}{3}\overrightarrow{BC}$ This mark can be implied by a correct (unsimplified) vector using their $\overrightarrow{OB}$ and their $\overrightarrow{BC}$.
	A1ft	For the correct simplified using their $\overrightarrow{OM}$ in terms of a single **a** and **b** only. $\overrightarrow{OM} = \frac{\boldsymbol{a}}{3} + \frac{5\boldsymbol{b}}{3}$ or $\frac{1}{3}(\boldsymbol{a} + 5\boldsymbol{b})$
(d)	M1 M1	M1 for one correct statement of route for $\overrightarrow{OY}$ M1 for second correct statement of route for $\overrightarrow{OY}$ $\overrightarrow{OY} = \mu\overrightarrow{OM}$ (or any other variable in place of μ) $\overrightarrow{OY} = \left(\overrightarrow{OA} + \overrightarrow{AY}\right) = \mathbf{a} + \lambda\mathbf{b}$ (or any other variable in place of λ, provided this is different to their μ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$
ALT – use of $\overrightarrow{BY}$ route, use also for $\overrightarrow{AY}$ route		
	M1 M1	M1 for one correct statement of route for $\overrightarrow{BY}$ M1 for second correct statement of route for $\overrightarrow{BY}$ $\overrightarrow{BY} = \lambda\overrightarrow{AB} = \lambda\mathbf{b}$ (or any other variable in place of λ) $\overrightarrow{BY} = \overrightarrow{BO} + \overrightarrow{OY} = \overrightarrow{BO} + \mu\overrightarrow{OM} = -(\boldsymbol{a} + \mathbf{b}) + \mu\left(\frac{1}{3}\mathbf{a} + \frac{5}{3}\boldsymbol{b}\right)$ $= \left(-1 + \frac{1}{3}\mu\right)\mathbf{a} + \left(-1 + \frac{5}{3}\mu\right)\mathbf{b}$ (or any other variable in place of μ provided this is different to their λ). Allow use of their vectors from earlier parts of the question.
	M1	For equating their coefficients of **a** and **b** to obtain two equations. Mark intent – one must be correct, condone slips in second.
	M1	Solving their simultaneous equations by any method. Only the value for their λ is required for this mark.
	A1	For $AB:BY = 1:4$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 17 of 20 · Question 8 part d · 5 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

(e)	**M1**	For use of the **correct** formula for area of a triangle with $60°$ and correct value of $\sin 60°$ $10 = \frac{1}{2}ab\sin 60° = ab \times \frac{\sqrt{3}}{4}$
	A1	For correct expression for ab or a $ab = \frac{40}{\sqrt{3}}$ or $a = \frac{40}{b\sqrt{3}}$
	dM1	For use of the **correct** formula for area of a triangle with $120°$ and attempt to substitute for ab or a. $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times '\frac{40}{b\sqrt{3}}' \times '5'b\sin 120°$ or $\text{Area} = \frac{1}{2}a \times '5'b\sin 120° = \frac{1}{2} \times ab \times '5'\sin 120° = \frac{1}{2} \times '\frac{40}{\sqrt{3}}' \times '5'\sin 120°$ Dependent on the first M awarded. Allow use of their λ from part (d).
	A1	For the correct area Area $= 50$
ALT – use of ratios of areas		
	M1	For use of their ratio $AB:BY$ to write an equation linking area OAY and area OAB $\frac{\text{Area } OAY}{\text{Area } OAB} = \frac{\frac{1}{2}\times h\times(1+'4')}{\frac{1}{2}\times h\times 1} = \frac{'5'}{1}$
	A1	For correct relationship between area OAY and area OAB
	dM1	For a correct method to find the area of OAB Dependent on first M mark being awarded.
	A1	For the correct area Area $= 50$

Type MS · Where lines cross: two routes and comparing coefficients

Answer 17 of 20 · Question 8 part d · 5 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 18 of 20 · Question 8 part c · 5 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

8 (a) (i)	$\overrightarrow{(AB)} = \overrightarrow{AO} + \overrightarrow{OB}$ $(= -4\mathbf{a} - 5\mathbf{b} + 8\mathbf{a} - \mathbf{b}) = 4\mathbf{a} - 6\mathbf{b}$	M1A1 [2]
(ii)	$\left\{ \left\| \overrightarrow{AB} \right\| = \sqrt{52} \right\} = 2\sqrt{13}$	B1 [1]
(b)	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2} \left(= \sqrt{41} \right)$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2} \left(= \sqrt{65} \right)$	M1
	$\left(\sqrt{52}\right)^2 = \left(\sqrt{41}\right)^2 + \left(\sqrt{65}\right)^2 - 2 \times \sqrt{41} \times \sqrt{65} \cos(AOB)$	M1
	$\left\{ \text{angle } AOB = \cos^{-1}\left(\frac{41 + 65 - 52}{2 \times \sqrt{41} \times \sqrt{65}} \right) = 58.465... \right\}$	
	Area triangle $AOB = \frac{1}{2} \times "\sqrt{41}" \times "\sqrt{65}" \times \sin "58.465..."$	M1
	$= 22$ awrt	A1 [4]
ALT 1	Area $= \frac{1}{2} \begin{vmatrix} 0 & 4 & 8 & 0 \\ 0 & 5 & -1 & 0 \end{vmatrix}$ oe	M1
	$= \frac{1}{2} \left\| (0 \times 5 + 4 \times (-1) + 8 \times 0) - (0 \times 4 + 5 \times 8 + (-1) \times 0) \right\|$ oe	M1
	$\left\{ = \frac{1}{2} \left| -44 \right| \right\} = 22$	M1A1 [4]
ALT 2	Use Heron's formula	
	$\left\| \overrightarrow{OA} \right\| = \sqrt{4^2 + 5^2} \left(= \sqrt{41} \right)$ *or* $\left\| \overrightarrow{OB} \right\| = \sqrt{8^2 + 1^2} \left(= \sqrt{65} \right)$	M1
	$p = \frac{"\sqrt{41}" + "\sqrt{65}" + "\sqrt{52}"}{2}$	M1
	Area $= \sqrt{p(p - "\sqrt{41}") \times (p - "\sqrt{65}") \times (p - "\sqrt{52}")}$	M1
	$= 22$ exact	A1 [4]
(c)	$\overrightarrow{OC} = \lambda \overrightarrow{OD} \left\{ = \lambda(15\mathbf{a} + 10\mathbf{b}) \right\}$	M1
	$\overrightarrow{OC} = \overrightarrow{OA} + \mu \overrightarrow{AB} \left\{ = 4\mathbf{a} + 5\mathbf{b} + \mu(4\mathbf{a} - 6\mathbf{b}) = (4 + 4\mu)\mathbf{a} + (5 - 6\mu)\mathbf{b} \right\}$	M1
	$4 + 4\mu = 15\lambda$ *and* $5 - 6\mu = 10\lambda \Rightarrow \mu = ...$ *or* $\lambda = ...$	ddM1
	$\mu = \frac{7}{26}$ *or* $\lambda = \frac{22}{65}$	A1
	$\overrightarrow{OC} = \frac{66}{13}\mathbf{a} + \frac{44}{13}\mathbf{b}$ or $5\frac{1}{13}\mathbf{a} + 3\frac{5}{13}\mathbf{b}$ cao	A1 [5]
		Total 12 marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 18 of 20 · Question 8 part c · 5 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)(i)	M1	For a correct vector path to find $\overrightarrow{AB}$ or $\overrightarrow{BA}$ or for $\pm(-4\mathbf{a}-5\mathbf{b}+8\mathbf{a}-\mathbf{b})$
	A1	For the correct vector $4\mathbf{a}-6\mathbf{b}$
(ii)	B1	For the correct exact simplest magnitude $2\sqrt{13}$
(b)	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For correct substitution into the cosine rule. e.g. $(\sqrt{65})^2=(\sqrt{41})^2+(\sqrt{52})^2-2\times\sqrt{41}\times\sqrt{52}\cos(OAB)$ or $(\sqrt{41})^2=(\sqrt{52})^2+(\sqrt{65})^2-2\times\sqrt{52}\times\sqrt{65}\cos(ABO)$
	M1	For correct substitution into the area of a triangle formula, use their values. e.g. $\frac{1}{2}\times\text{"}\sqrt{41}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}72.349...\text{"}$ or $\frac{1}{2}\times\text{"}\sqrt{65}\text{"}\times\text{"}\sqrt{52}\text{"}\times\sin\text{"}49.184...\text{"}$
	A1	awrt 22
ALT1	M1	For a correct statement for the area, must have $\frac{1}{2}$ (Start with any point, go clockwise or anticlockwise, but must repeat the first point at the end.)
	M1	For a correct evaluation of their determinant, ignore $\frac{1}{2}$
	M1	$\frac{1}{2}$ multiplies their determinant value, correct answer implies this mark.
	A1	Correct **exact** value 22, **not awrt**
ALT2	M1	For the correct simplified or unsimplified $\lvert\overrightarrow{OA}\rvert$ or $\lvert\overrightarrow{OB}\rvert$
	M1	For finding half of the perimeter
	M1	Substitute the perimeter and 3 lengths to a correct Heron's formula
	A1	Correct **exact** value 22, **not awrt**
(c)	M1	For $\overrightarrow{OC}=\lambda\overrightarrow{OD}$ (can be implied in their later work)
	M1	For correct vector statement $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}$ oe Can be implied for later substitution of their vectors **Please note**: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{BA}$ or $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{AB}$ is also correct, provided they have found a negative value of μ
	ddM1	Equating components and reaching a value for either of their parameters. Dependent on both previous method marks
	A1	Either correct value of their parameters. They do not need to work out both values. Note: $\overrightarrow{OC}=\overrightarrow{OA}+\mu\overrightarrow{AB}\Rightarrow\mu=\frac{7}{26}$, $\overrightarrow{OC}=\overrightarrow{OB}+\mu\overrightarrow{BA}\Rightarrow\mu=\frac{19}{26}$
	A1	Correct vector in simplified expression

Type MS · Where lines cross: two routes and comparing coefficients

Answer 18 of 20 · Question 8 part c · 5 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 19 of 20 · Question 11 part b · 9 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

Question number		
11 (a)	$\mathbf{b}-\mathbf{a}$	B1 (1)
(b)	$\overrightarrow{OZ}=\overrightarrow{OB}+\lambda\overrightarrow{BX}(=\mathbf{b}+\lambda(-\mathbf{b}+2\mathbf{a}))$	M1
	$=(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}$	A1
	$\overrightarrow{OZ}=\overrightarrow{OA}+\mu\overrightarrow{AY}(=\mathbf{a}+\mu(-\mathbf{a}+3\mathbf{b}))$	M1
	$=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	A1
	$(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	ddM1
	$2\lambda=1-\mu$ $3\mu=1-\lambda$	A1
	$3(1-2\lambda)=1-\lambda$ or $2(1-3\mu)=1-\mu$	M1
	$\lambda=\frac{2}{5}$ or $\mu=\frac{1}{5}$	A1
	$\overrightarrow{OZ}=\frac{1}{5}(4\mathbf{a}+3\mathbf{b})$ See notes regarding alternatives	A1 (9)
(c)	$\overrightarrow{OM}=p"\frac{1}{5}(4\mathbf{a}+3\mathbf{b})"$ and $\overrightarrow{OM}=2\mathbf{a}+q(-2\mathbf{a}+3\mathbf{b})$	M1
	$\frac{4p}{5}=2-2q$ and $\frac{3p}{5}=3q$ (Solving these equations leads to $p=\frac{5}{3}$)	M1
	$\overrightarrow{OM}=\frac{1}{3}(4\mathbf{a}+3\mathbf{b})$	A1 (3) **[13]**

Type MS · Where lines cross: two routes and comparing coefficients

Answer 19 of 20 · Question 11 part b · 9 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the indicated vector
(b)	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	ddM1	Equates their 2 vectors – this mark may be implicit in the candidate equating the two components of their 2 vectors, **dependent on the first two method marks.**
	A1	Correct equations as shown
	ddM1	Full and correct method to solve their two simultaneous equations, either by substitution as shown or by elimination. There must be no errors in the method to eliminate λ or μ, **dependent on the first two method marks.**
	A1	Correct value for λ or μ
	A1	Correct vector.
	There are a number of alternatives for part b, all marked in the same way. Examples: ALT1 $\mu\overrightarrow{XB}$ and $\lambda\overrightarrow{AY}$ (must use to get $\overrightarrow{AB}$ for 2nd A1) equate 2 vectors for $\overrightarrow{AB}$ ALT2 $\overrightarrow{AZ} = \overrightarrow{AX} + \mu\overrightarrow{XB}$ M1A1 $\overrightarrow{AZ} = \lambda\overrightarrow{AY}$ M1A1 equate 2 vectors for $\overrightarrow{AZ}$ ddM1 The A1 ddM1 A1 A1 following these marks should all be marked in the same way as the main mark scheme.	
(c)	M1	For the two correct vectors shown, allow use of their $\overrightarrow{OZ}$
	dM1	Correctly equating the components of their vectors for $\overrightarrow{OZ}$ and arriving at a value for p or q
	A1	For the correct vector, as shown.
	Can also be done using other vectors eg finding two alternatives for $\overrightarrow{OM}$ Mark in the same way as main scheme.	

Type MS · Where lines cross: two routes and comparing coefficients

Answer 19 of 20 · Question 11 part b · 9 marks

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Where lines cross: two routes and comparing coefficients

Answer 20 of 20 · Question 11 part c · 3 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

Question number		
11 (a)	$\mathbf{b}-\mathbf{a}$	B1 (1)
(b)	$\overrightarrow{OZ}=\overrightarrow{OB}+\lambda\overrightarrow{BX}(=\mathbf{b}+\lambda(-\mathbf{b}+2\mathbf{a}))$	M1
	$=(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}$	A1
	$\overrightarrow{OZ}=\overrightarrow{OA}+\mu\overrightarrow{AY}(=\mathbf{a}+\mu(-\mathbf{a}+3\mathbf{b}))$	M1
	$=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	A1
	$(1-\lambda)\mathbf{b}+2\lambda\mathbf{a}=(1-\mu)\mathbf{a}+3\mu\mathbf{b}$	ddM1
	$2\lambda=1-\mu$ $3\mu=1-\lambda$	A1
	$3(1-2\lambda)=1-\lambda$ or $2(1-3\mu)=1-\mu$	M1
	$\lambda=\frac{2}{5}$ or $\mu=\frac{1}{5}$	A1
	$\overrightarrow{OZ}=\frac{1}{5}(4\mathbf{a}+3\mathbf{b})$ See notes regarding alternatives	A1 (9)
(c)	$\overrightarrow{OM}=p"\frac{1}{5}(4\mathbf{a}+3\mathbf{b})"$ and $\overrightarrow{OM}=2\mathbf{a}+q(-2\mathbf{a}+3\mathbf{b})$	M1
	$\frac{4p}{5}=2-2q$ and $\frac{3p}{5}=3q$ (Solving these equations leads to $p=\frac{5}{3}$)	M1
	$\overrightarrow{OM}=\frac{1}{3}(4\mathbf{a}+3\mathbf{b})$	A1 (3)
		[13]

Type MS · Where lines cross: two routes and comparing coefficients

Answer 20 of 20 · Question 11 part c · 3 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	For the indicated vector
(b)	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	M1	For any correctly written vector path, must include a parameter
	A1	For the vector shown
	ddM1	Equates their 2 vectors – this mark may be implicit in the candidate equating the two components of their 2 vectors, **dependent on the first two method marks.**
	A1	Correct equations as shown
	ddM1	Full and correct method to solve their two simultaneous equations, either by substitution as shown or by elimination. There must be no errors in the method to eliminate λ or μ, **dependent on the first two method marks.**
	A1	Correct value for λ or μ
	A1	Correct vector.
	There are a number of alternatives for part b, all marked in the same way. Examples: ALT1 $\mu\overrightarrow{XB}$ and $\lambda\overrightarrow{AY}$ (must use to get $\overrightarrow{AB}$ for 2nd A1) equate 2 vectors for $\overrightarrow{AB}$ ALT2 $\overrightarrow{AZ} = \overrightarrow{AX} + \mu\overrightarrow{XB}$ M1A1 $\overrightarrow{AZ} = \lambda\overrightarrow{AY}$ M1A1 equate 2 vectors for $\overrightarrow{AZ}$ ddM1 The A1 ddM1 A1 A1 following these marks should all be marked in the same way as the main mark scheme.	
(c)	M1	For the two correct vectors shown, allow use of their $\overrightarrow{OZ}$
	dM1	Correctly equating the components of their vectors for $\overrightarrow{OZ}$ and arriving at a value for p or q
	A1	For the correct vector, as shown.
	Can also be done using other vectors eg finding two alternatives for $\overrightarrow{OM}$ Mark in the same way as main scheme.	

Type MS · Where lines cross: two routes and comparing coefficients

Answer 20 of 20 · Question 11 part c · 3 marks

2020 NOV · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks